Conjecture on the Number of Linearly Independent Relations of Ihara–Kaneko–Zagier Derivation Relation: An Euler-type Pentagonal Formula

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October 19, 2026 – Szeged

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1. Generalized Pentagonal Numbers and the $\omega(n)$ Function

Let $P_0, P_1, P_2, P_3, P_4, P_5, P_6, P_7, \ldots = 0, 1, 2, 5, 7, 12, 15, 22, \ldots$ denote the generalized pentagonal numbers OEIS:A001318, and let $\omega(k)$= A010815(k), defined by

$$\omega(k) = \begin{cases} 1 & \text{if } k = 0; \\ (-1)^m & \text{if } k = \frac{3m^2 \pm m}{2}; \\ 0 & \text{otherwise.} \end{cases}$$

Table 1. P_n and $\omega(n)$

n	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
P_n	0	1	2	5	7	12	15	22	26	35	40	51	57	70	77	92	100	117	126	145	155	176	187	210
$\omega(n)$	1	-1	-1	0	0	1	0	1	0	0	0	0	-1	0	0	-1	0	0	0	0	0	0	1	0

2. The Conjecture and Some Equivalent Formulations

For a given weight $n \geq 3$ let $\mathscr{L}(n)$ denote the the number of linearly independent relations of Ihara-Kaneko-Zagier derivation relation. In [3, 5] the authors have tabulated the values of $\mathscr{L}(n)$ calculated using the Risa/Asir and Mathematica general computer algebra system for $n = 3$ to $n = 15$.

Table 2. Number of Independent Derivation Relations for MZVs

n	3	4	5	6	7	8	9	10	11	12	13	14	15	16*	17*
$\mathscr{L}(n)$	1	2	5	10	22	44	90	181	363	727	1456	2912	5825	11652*	23304*

* Calculated with a JavaScript application of my own development.[1]

Conjecture 1. *For all weight $n \geq 3$ the number of linearly independent relations of Ihara-Kaneko-Zagier derivation relation can be calculated by the following Euler-type pentagonal formula:*

$$\mathscr{L}(n) = 2^{n-2-P_1} + 2^{n-2-P_2} - 2^{n-2-P_3} - 2^{n-2-P_4} \pm \cdots - \omega(n-1) - 1 = -\sum_{k=1}^{n-2} \omega(k)\, 2^{n-2-k} - \omega(n-1) - 1.$$

The generating function of $\mathscr{L}(n)$ is given by:

$$F(x) = \frac{x^2}{x-1} + \frac{x(x-1)}{2x-1} \prod_{k \geq 1} (1 - x^k),$$

and $\mathscr{L}(n)$ satisfies the following recurrence relation for all $n \geq 3$:

$$\mathscr{L}(n+1) = 2 \cdot \mathscr{L}(n) + 1 - \omega(n) + \omega(n-1).$$

The calculations below verify that Conjecture 1. holds for $n = 3$ through $n = 17$:

$\mathscr{L}(3) = 2^{1-1} - \omega(2) - 1 = 2^0 + 1 - 1 - 1 = 1 + 1 - 1 - 1 = \mathbf{1}$

$\mathscr{L}(4) = 2^{2-1} + 2^{2-2} - \omega(3) - 1 = 2^1 + 2^0 - 0 - 1 = 2 + 1 - 1 = \mathbf{2}$

$\mathscr{L}(5) = 2^{3-1} + 2^{3-2} - \omega(4) - 1 = 2^2 + 2^1 - 0 - 1 = 4 + 2 - 1 = \mathbf{5}$

$\mathscr{L}(6) = 2^{4-1} + 2^{4-2} - \omega(5) - 1 = 2^3 + 2^2 - 1 - 1 = 8 + 4 - 1 - 1 = \mathbf{10}$

$\mathscr{L}(7) = 2^{5-1} + 2^{5-2} - 2^{5-5} - \omega(6) - 1 = 2^4 + 2^3 - 2^0 - 0 - 1 = 16 + 8 - 1 - 1 = \mathbf{22}$

[1] On page https://tamas-tech.github.io

$\mathscr{L}(8) = 2^{6-1} + 2^{6-2} - 2^{6-5} - \omega(7) - 1 = 2^5 + 2^4 - 2^1 - 1 - 1 = 32 + 16 - 2 - 1 - 1 = \mathbf{44}$

$\mathscr{L}(9) = 2^{7-1} + 2^{7-2} - 2^{7-5} - 2^{7-7} - \omega(8) - 1 = 2^6 + 2^5 - 2^2 - 2^0 - 0 - 1 = 64 + 32 - 4 - 1 - 1 = \mathbf{90}$

$\mathscr{L}(10) = 2^{8-1} + 2^{8-2} - 2^{8-5} - 2^{8-7} - \omega(9) - 1 = 2^7 + 2^6 - 2^3 - 2^1 - 0 - 1 = 128 + 64 - 8 - 2 - 1 = \mathbf{181}$

$\mathscr{L}(11) = 2^{9-1} + 2^{9-2} - 2^{9-5} - 2^{9-7} - \omega(10) - 1 = 2^8 + 2^7 - 2^4 - 2^2 - 0 - 1 = 256 + 128 - 16 - 4 - 1 = \mathbf{363}$

$\mathscr{L}(12) = 2^{10-1} + 2^{10-2} - 2^{10-5} - 2^{10-7} - \omega(11) - 1 = 2^9 + 2^8 - 2^5 - 2^3 - 0 - 1 = 512 + 256 - 32 - 8 - 1 = \mathbf{727}$

$\mathscr{L}(13) = 2^{11-1} + 2^{11-2} - 2^{11-5} - 2^{11-7} - \omega(12) - 1 = 2^{10} + 2^9 - 2^6 - 2^4 + 1 - 1 = 1024 + 512 - 64 - 16 = \mathbf{1456}$

$\mathscr{L}(14) = 2^{12-1} + 2^{12-2} - 2^{12-5} - 2^{12-7} + 2^{12-12} - \omega(13) - 1 = 2^{11} + 2^{10} - 2^7 - 2^5 + 2^0 - 0 - 1 = 2048 + 1024 - 128 - 32 + 1 - 1 = \mathbf{2912}$

$\mathscr{L}(15) = 2^{13-1} + 2^{13-2} - 2^{13-5} - 2^{13-7} + 2^{13-12} - \omega(14) - 1 = 2^{12} + 2^{11} - 2^8 - 2^6 + 2^1 - 0 - 1 = 4096 + 2048 - 256 - 64 + 2 - 1 = \mathbf{5825}$

$\mathscr{L}(16) = 2^{14-1} + 2^{14-2} - 2^{14-5} - 2^{14-7} + 2^{14-12} - \omega(15) - 1 = 2^{13} + 2^{12} - 2^9 - 2^7 + 2^2 + 1 - 1 = 8192 + 4096 - 512 - 128 + 4 = \mathbf{11652}$

$\mathscr{L}(17) = 2^{15-1} + 2^{15-2} - 2^{15-5} - 2^{15-7} + 2^{15-12} + 2^{15-15} - \omega(16) - 1 = 2^{14} + 2^{13} - 2^{10} - 2^8 + 2^3 + 2^0 - 0 - 1 =$

$= 16384 + 8192 - 1024 - 256 + 8 + 1 - 1 = \mathbf{23304}$

The first equivalent version reveals a simple connection between $\mathscr{L}(n)$ and pic(n) =A178841(n) the number of pure inverting compositions of n.

Conjecture 2. *For all weight $n \geq 3$ the number of linearly independent relations of Ihara-Kaneko-Zagier derivation relation can be calculated by the following formula:*

$$\mathscr{L}(n) = 2^{n-2} - 1 - pic(n-1)$$

The calculations summarized in the table below verify that Conjecture 2. holds for $n = 3$ through $n = 17$:

Table 3. $\mathscr{L}(n) = 2^{n-2} - 1 - \text{pic}(n-1)$

n	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$2^{n-2}-1$	1	3	7	15	31	63	127	255	511	1023	2047	4095	8191	16383	32767
pic$(n-1)$	0	1	2	5	9	19	37	74	148	296	591	1183	2366	4731	9463
$2^{n-2}-1-\text{pic}(n-1)$	1	2	5	10	22	44	90	181	363	727	1456	2912	5825	11652	23304

Let $c(n)$ =A152537(n). The second equivalent reformulation highlights a deeper algebraic relationship between $c(n)$ and pic(n).

Conjecture 3. *For all weight $n \geq 3$ the number of linearly independent relations of Ihara-Kaneko-Zagier derivation relation can be calculated by the following formula:*

$$\mathscr{L}(n) = 2^{n-2} - 1 - \omega(n-1) - c(n-2)$$

The calculations summarized in the table below verify that Conjecture 3. holds for $n = 3$ through $n = 17$:

Table 4. $\mathscr{L}(n) = 2^{n-2} - 1 - \omega(n-1) - c(n-2)$

n	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$2^{n-2}-1$	1	3	7	15	31	63	127	255	511	1023	2047	4095	8191	16383	32767
$\omega(n-1)$	-1	0	0	1	0	1	0	0	0	0	-1	0	0	-1	0
$c(n-2)$	1	1	2	4	9	18	37	74	148	296	592	1183	2366	4732	9463
$2^{n-2}-1-\omega(n-1)-c(n-2)$	1	2	5	10	22	44	90	181	363	727	1456	2912	5825	11652	23304

3. Empirical and Theoretical Results Leading to Conjecture 1

Before proceeding, we establish some terminology and notation that will be used throughout the remainder of this work.

Le $\mathfrak{H}_n := \{w \in \mathfrak{H}, |w| = n\}$ denote the set of words of weight n. We recall that the set of admissible words is given by $\mathcal{A} = x\mathfrak{H}y$, and we denote the subset of admissible words of weight n by $\mathcal{A}_n := x\mathfrak{H}_{n-2}y$. For any admissible word $w \in \mathcal{A}$, we define its binary value, denoted by bin(w) , as the unique even natural number whose binary representation yields the word w upon replacing each digit 1 with x and each digit 0 with y. Equivalently, if a word is written as $w = u_1u_2\ldots u_k$, where $u_i \in \{x, y\}$ then

$$\text{bin}(w) = \sum_{i=1}^{k} c_i 2^{k-i}, \text{ where } c_i = 1 \text{ if } u_i = x, \text{ and } c_i = 0 \text{ if } u_i = y.$$

For example, consider the admissible word $w = xxyy$. Replacing each x with 1 and y with 0 gives the binary string 1100_2. Converting this to base 10, we obtain: $\text{bin}(xxyy) = 1 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 0 \cdot 2^0 = 8 + 4 = 12$.

We will use a graded lexicographic total order on the words of $\mathfrak{H}$, defined as follows. If u and v are words with $|u| < |v|$, set $u < v$. If $|u| = |v|$ and $u = wyw_2$ and $v = wxw_3$, for any words w,w_1, and w_2, set $u < v$. It is easy to see that the order on $\mathcal{A}$ induced by the mapping $\text{bin}(\cdot)$ coincides with the graded lexicographic order restricted to $\mathcal{A}$. The words of degree at most 3 are thus ordered as follows:

$$1 < y < x < yy < yx < xy < xx < yyy < yyx < yxy < yxx < xyy < xyx < xxy < xxx$$

For any subset $V \subseteq \mathfrak{H}$ and any integer $m \geq 1$, let $\partial_m(V)$ denote the set $\{\partial_m(w) : w \in V\}$.

Conjecture 1 is based on the following analysis of the rank growth, which chronologically preceded the formulation of Conjecture 2.

For a fixed $n \geq 3$, we will construct a full row-rank matrix $\mathbf{M}_n$ from the derivatives $\partial_{n-|w|}(w)$ of the elements w in the subset

$$\mathcal{B}_n = \mathcal{A}_{n-1} \cup x\mathcal{A}_{n-3} \cup x\mathcal{A}_{n-2} \cup \cdots \cup x\mathcal{A}_2 \cup \mathcal{A}_2 \subset h$$

$\left(|\mathcal{B}_n| = 2^{n-3} + (2^{n-5} + 2^{n-6} + \cdots + 2^0) + 1 = 3 \cdot 2^{n-4}\right)$ according to the following procedure:

The columns of $\mathbf{M}_n$ are indexed by the elements of the set of admissible words of $\mathcal{A}_n$, ordered ascendingly with respect to the graded lexicographic order.

(1) Let $w = x^{n-2}y$ be the maximal element of the set $\mathcal{A}_{n-1}$. The first row of the matrix $\mathbf{M}_n$ is defined as the coordinate vector of the derivative $\partial_1(w) \in \partial_1(\mathcal{A}_{n-1}) \subset \text{Span}(\mathcal{A}_n)$ relative to the ordered basis $(\mathcal{A}_n, <)$.

(2) We then proceed iteratively through the remaining elements of A_{n-1} in descending order. For each element, we compute the coordinate vector of its derivative and insert it as a new row in $\mathbf{M}_n$. *Crucially, throughout this process, we ensure that if a candidate row can be expressed as a linear combination of the preceding rows, it is discarded.*

(3) This procedure is then repeated analogously for the elements of the remaining subsets in $\mathcal{B}_n$ (namely $x\mathcal{A}_{n-3}, x\mathcal{A}_{n-2}, \ldots, x\mathcal{A}_2$, and $\mathcal{A}_2$ in the given order), examining their corresponding derivatives in descending order within each subset.

Constructing the matrix via this elimination process, it is evident that the resulting matrix $\mathbf{M}_n$ will have full row-rank. At the end of the procedure, we count how many derivatives originating from the subsets $\mathcal{A}_{n-1}, x\mathcal{A}_{n-3}, x\mathcal{A}_{n-2}, \ldots, x\mathcal{A}_2$, and $\mathcal{A}_2$ remain as rows in the matrix. These quantities will be denoted by $r_{0,n-2}, r_{1,n-2}, r_{2,n-2}, \ldots, r_{n-3,n-2}$ respectively, and they will constitute the entries of the $(n-2)$-th column of an upper triangular matrix $\mathbf{R}$, whose rows and columns are indexed starting from 0.

Since our JavaScript application, which includes the implementation of the above procedure, generates a clear graphical layout for smaller values of n, the visual output representing the matrix $\mathbf{M}_6$ for $n = 6$ is included below as an illustrative example.

μ(w)	ρ	D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
w	rank	Delete	xyyyyy	xyyyxy	xyyxyy	xyyxxy	xyxyyy	xyxyxy	xyxxyy	xyxxxy	xxyyyy	xxyyxy	xxyxyy	xxyxxy	xxxyyy	xxxyxy	xxxxyy	xxxxxy
∂_1^1(xxxxy)	1	✗	0	0	0	0	0	0	0	1	0	0	0	1	0	1	1	-1
∂_1^1(xxxyy)	2	✗	0	0	0	0	0	0	1	0	0	0	1	0	1	-1	-1	0
∂_1^1(xxyxy)	3	✗	0	0	0	0	0	1	0	0	0	1	1	-1	0	-1	0	0
∂_1^1(xxyyy)	4	✗	0	0	0	0	1	0	0	0	1	-1	-1	0	-1	0	0	0
∂_1^1(xyxxy)	5	✗	0	0	0	1	0	1	1	-1	0	0	0	-1	0	0	0	0
∂_1^1(xyxyy)	6	✗	0	0	1	0	1	-1	-1	0	0	0	-1	0	0	0	0	0
∂_1^1(xyyxy)	7	✗	0	1	1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0
∂_1^1(xyyyy)	8	✗	1	-1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0	0
∂_2^1(xxxy)	9	✗	0	0	0	1	0	0	0	0	0	1	0	1	1	1	0	-1
∂_2^1(xxyy)	10	✗	0	0	1	0	0	0	0	0	1	0	0	-1	0	0	-1	0

For $n = 6$, the generating set is given by $\mathcal{B}_6 = \mathcal{A}_5 \cup x\mathcal{A}_3 \cup x\mathcal{A}_2 \cup \mathcal{A}_2$. As clearly observed from the visual output, all $2^3 = 8$ elements of $\partial_1(\mathcal{A}_5)$ and all $2^1 = 2$ elements of $\partial_2(x\mathcal{A}_3)$ remain in the matrix. Conversely, the elements of $\partial_3(x\mathcal{A}_2) = \{\partial_3(xxy)\}$

and $\partial_4(\mathcal{A}_2) = \{\partial_4(xy)\}$ can already be expressed as linear combinations of the preceding rows and are thus discarded. By definition, this yields $r_{0,4} = 8$, $r_{1,4} = 2$, and $r_{2,4} = r_{3,4} = 0$.

To provide an additional illustration, let us consider the output generated for the case $n = 7$.

μ(w)	ρ	D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
w	rank	Delete	xyyyyyy	xyyyyxy	xyyyxyy	xyyyxxy	xyyxyyy	xyyxyxy	xyyxxyy	xyyxxxy	xyxyyyy	xyxyyxy	xyxyxyy	xyxyxxy	xyxxyyy	xyxxyxy	xyxxxyy	xyxxxxy	xxyyyyy	xxyyyyx	xxyyyxy	xxyyxyy	xxyyxyx	xxyyxxy	xxyxyyy	xxyxyyx	xxyxyxy	xxyxxyy	xxyxxxy	xxxyyyy	xxxyyyx	xxxyyxy	xxxyxyy	xxxyxxy	xxxxyyy	xxxxyyx	xxxxyxy	xxxxxyy	xxxxxxy
∂_1^1(xxxxxy)	1	✗	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	1	0	1	1	-1
∂_1^1(xxxxyy)	2	✗	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	1	0	1	-1	-1	0
∂_1^1(xxxyxy)	3	✗	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	1	1	-1	0	-1	0	0
∂_1^1(xxxyyy)	4	✗	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	1	-1	-1	0	-1	0	0	0
∂_1^1(xxyxxy)	5	✗	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	1	1	-1	0	0	0	-1	0	0	0	0
∂_1^1(xxyxyy)	6	✗	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	1	-1	-1	0	0	0	-1	0	0	0	0	0
∂_1^1(xxyyxy)	7	✗	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0
∂_1^1(xxyyyy)	8	✗	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	-1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0	0
∂_1^1(xyxxxy)	9	✗	0	0	0	0	0	0	0	1	0	0	0	1	0	1	1	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0
∂_1^1(xyxxyy)	10	✗	0	0	0	0	0	0	1	0	0	0	1	0	1	-1	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0
∂_1^1(xyxyxy)	11	✗	0	0	0	0	0	1	0	0	0	1	1	-1	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0
∂_1^1(xyxyyy)	12	✗	0	0	0	0	1	0	0	0	1	-1	-1	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0
∂_1^1(xyyxxy)	13	✗	0	0	0	1	0	1	1	-1	0	0	0	-1	0	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0
∂_1^1(xyyxyy)	14	✗	0	0	1	0	1	-1	-1	0	0	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0	0
∂_1^1(xyyyxy)	15	✗	0	1	1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
∂_1^1(xyyyyy)	16	✗	1	-1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
∂_2^1(xxxxy)	17	✗	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	1	0	1	1	1	0	-1
∂_2^1(xxxyy)	18	✗	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	1	0	0	-1	0	0	-1	0
∂_2^1(xxyxy)	19	✗	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	1	0	-1	0	0	0	0	0	-1	0	0
∂_2^1(xxyyy)	20	✗	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	-1	-1	0	0	-1	0	0	0	0	0	-1	0	0	0
∂_3^1(xxxy)	21	✗	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	1	0	1	0	0	1	1	1	1	0	0	0	-1
∂_3^1(xxyy)	22	✗	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	1	0	0	-1	-1	-1	0	0	0	0	0	0	-1	0

From this output, it can be observed that all 16, 4, and 2 elements of $\partial_1(\mathcal{A}_6)$, $\partial_2(x\mathcal{A}_4)$, and $\partial_3(x\mathcal{A}_3)$, respectively, remain in the matrix, whereas all elements of $\partial_4(x\mathcal{A}_2)$ and $\partial_5(\mathcal{A}_2)$ are eliminated. By definition, this yields $r_{0,5} = 16$, $r_{1,5} = 4$, $r_{2,5} = 2$, and $r_{3,5} = r_{4,5} = 0$.

Next, we present the empirical results obtained for $r_{i,j}$ from $n = 2$ to $n = 17$, collected into the table below:

	$n =$	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	$i \setminus j$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
∂_1	0	1	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	⋯
∂_2	1			0	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	⋯
∂_3	2				0	0	2	3	6	12	24	48	96	192	384	768	1536	⋯
∂_4	3					0	0	1	3	6	10	21	42	84	168	336	672	⋯
∂_5	4						0	0	1	2	6	10	20	39	78	158	315	⋯
∂_6	5							0	0	1	2	5	10	20	38	76	152	⋯
∂_7	6								0	0	1	2	5	9	20	38	75	⋯
∂_8	7									0	0	1	2	5	9	19	38	⋯
∂_9	8										0	0	1	2	5	9	19	⋯
∂_{10}	9											0	0	1	2	5	9	⋯
∂_{11}	10												0	0	1	2	5	⋯
∂_{12}	11													0	0	1	2	⋯
∂_{13}	12														0	0	1	⋯
∂_{14}	13															0	0	⋯
∂_{15}	14																0	⋯
∂_{16}	15																	⋯⋯

The most significant observation regarding the resulting table is that, putting aside the first few entries of each row, the values follow a clearly discernible recurrence relation. Specifically, the j-th entry of the i-th row can be obtained as the difference between the $(j-1)$-th entry of the $(i-1)$-th row and the $(j-i-1)$-th entry of the $(i-1)$-th row.

Formally, for sufficiently large indices, the entries satisfy the following recurrence: $r_{i,j} = r_{i-1,j-1} - r_{i-1,j-i-1}$. To illustrate this pattern, consider the following examples taken directly from the empirical data:

- For $i = 4$ and $j = 12$ (corresponding to ∂_5 at $n = 14$), we have $39 = 42 - 3$.
- For $i = 3$ and $j = 13$ (corresponding to ∂_4 at $n = 15$), we have $168 = 192 - 24$.

By applying this recurrence relation backward, we can determine the exact initial values required at the beginning of each row for the relation to hold globally across the entire table. Although such a backward extension could potentially lead to a contradiction, remarkably, no such inconsistency arises. Specifically, if we set all entries on the main diagonal to 1 and all entries strictly below the main diagonal to 0, the recurrence relation is satisfied consistently throughout the entire table. The resulting extended table is presented below:

$i \setminus j$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
0	**1**	**1**	**2**	**4**	**8**	**16**	**32**	**64**	**128**	**256**	**512**	**1024**	**2048**	**4096**	**8192**	**16384**	$\cdots$
1	0	1	0	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	$\cdots$
2	0	0	1	0	0	2	3	6	12	24	48	96	192	384	768	1536	$\cdots$
3	0	0	0	1	0	0	1	3	6	10	21	42	84	168	336	672	$\cdots$
4	0	0	0	0	1	0	0	1	2	6	10	20	39	78	158	315	$\cdots$
5	0	0	0	0	0	1	0	0	1	2	5	10	20	38	76	152	$\cdots$
6	0	0	0	0	0	0	1	0	0	1	2	5	9	20	38	75	$\cdots$
7	0	0	0	0	0	0	0	1	0	0	1	2	5	9	19	38	$\cdots$
8	0	0	0	0	0	0	0	0	1	0	0	1	2	5	9	19	$\cdots$
9	0	0	0	0	0	0	0	0	0	1	0	0	1	2	5	9	$\cdots$
10	0	0	0	0	0	0	0	0	0	0	1	0	0	1	2	5	$\cdots$
11	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	2	$\cdots$
12	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	$\cdots$
13	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	$\cdots$
14	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	$\cdots$
15	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$
$\sum_{j=1}^{i} a_{i,j}$	1	**2**	**3**	**6**	**11**	**23**	**45**	**91**	**182**	**364**	**728**	**1457**	**2913**	**5826**	**11653**	**23305**	$\cdots$

It naturally follows from the construction of the table that subtracting 1 from the sum of the entries in its j-th column yields exactly the values of $\mathscr{L}(j+2)$ for all $j \geq 1$.

A formalization of the recurrence relation observed in the table is given by:

$$r_{i,j} = \begin{cases} 0 & \text{if } i < 0 \text{ or } j < 0; \\ 1 & \text{if } i = j \text{ for } i = 0, 1, 2, 3, \dots; \\ 2^{\max(0,j-1)} & \text{if } i = 1 \text{ for } j = 0, 1, 2, 3, \dots; \\ a_{i-1,j-1} - a_{i-1,j-i-1} & \text{otherwise.} \end{cases}$$

The recurrence relation was investigated within a more general theoretical framework as a transformation. Instead of the initial values $2^{\max(0,j-1)}$ entered into the first row, let us consider an arbitrary arithmetic sequence $\boldsymbol{a} = (a(1), a(2), \dots)$, and let the column sums define the values of the transformed sequence $T(\boldsymbol{a}) = (T(a(1)), T(a(2)), \dots)$. During our thorough theoretical investigation, it soon became apparent that this transformation is remarkably versatile and well-suited for providing concise and clear explanations of the connections between various combinatorial results with a special emphasis on pentagonal formulas. Furthermore, since it is based on shifted differences and a summation, the transformation is inherently linear. What is essential for our purposes is that the transformation maps the sequence $\boldsymbol{a} = (1, 0, 0, 0, \dots)$ to the sequence $-\boldsymbol{\omega} = (1, 1, 0, 0, -1, 0, -1, 0, 0, 0, 0, 1, 0, \dots)$ (now, starting from 1 rather than 0 as done above). Consequently, by linearity, the transform of an arbitrary sequence $\boldsymbol{a} = (a_1, a_2, a_3, \dots)$ yields a pentagonal formula of the form $-\sum_{k=1}^{\infty} \omega(k) a_{n-k}$, which can be written more concisely as $T(\boldsymbol{a}) = -(a_1, a_2, \dots) * (\omega(1), \omega(2), \dots)$, where $*$ denotes the convolution of sequences. This, along with a few straightforward algebraic manipulations, directly recovers the pentagonal formula presented in Conjecture 1. Since the specific areas discussing this type of transformation lie beyond the current scope of our knowledge, we shall adopt the term *pentagonal transformation* for the remainder of this work. During the study of the transformation, it also became apparent that when transforming various sequences $\boldsymbol{a} = (a(1), a(2), \dots)$, it is not only the transformed sequence $T(\mathbf{a})$ itself that is of interest; the individual rows generated in the table during the underlying recurrence relation, as well as their corresponding generating functions, are also highly significant. In the table above leading to the values of $\mathscr{L}(j)$, for instance, it is clearly observable that

the first non-zero entries of successive rows coincide with the sequence $(\mathrm{pic}(0), \mathrm{pic}(1), \mathrm{pic}(2), \dots) = (1, 0, 0, 1, 2, 5, 9, 19, 37, \dots)$ for increasingly longer stretches. From this, together with certain relations to be discussed in Section 5, it will become completely evident that each row of the table contains parametric generalizations $(\mathrm{pic}_n(0), \mathrm{pic}_n(1), \mathrm{pic}_n(2), \dots)$ of the sequence $(\mathrm{pic}(0), \mathrm{pic}(1), \mathrm{pic}(2), \dots)$ corresponding to the parameter n of that row.

As for the proof of Conjecture 1, it suffices to demonstrate two conditions:

(1) for any $n \geq 3$, the vectors in the set $\partial_1(\mathcal{A}_{n-1})$ are linearly independent;
(2) the rank increments $r_{i,j}$ follow the aforementioned recurrence relation.

The first statement can be straightforwardly established without any difficulty, from which the corresponding power of two naturally follows. However, we have been unable to provide a proof for the second statement.

It is worth emphasizing that the credibility of Conjecture 1 is significantly reinforced by the fact that the consistency does not merely rest on the first fourteen terms of a single sequence. Rather, it is deeply embedded in the entire underlying table of rank increments, matching perfectly across the whole two-dimensional recursive structure.

4. Empirical and Theoretical Results Leading to Conjecture 2

The empirical results leading to Conjecture 2 can be obtained via the exact same elimination procedure described for Conjecture 1; however, rather than traversing the words in descending order, the matrix is iteratively expanded by examining the coordinate vectors of their derivatives in ascending order. This modification offers a distinct computational advantage: as can be readily verified, the coordinate vectors of the derivatives in the set

$$\mathcal{B}_n^* = \mathcal{B}_n \setminus \mathcal{A}_{n-1} = \partial_{n-2}(\mathcal{A}_2) \cup \partial_{n-3}(x\mathcal{A}_2) \cup \cdots \cup \partial_2(x\mathcal{A}_{n-3})$$

relative to the ordered basis $(\mathcal{A}_n, <)$ are guaranteed to form a linearly independent system of vectors. Consequently, for any fixed $n \geq 3$, the elimination procedure can be initiated immediately with the full row-rank matrix $\mathbf{N}_n$ whose rows are defined as the coordinate vectors of the derivatives contained in the set $\mathcal{B}_n^*$ without requiring any prior rank computation. This initial matrix $\mathbf{N}_n$ consists of exactly $1 + (2^0 + 2^1 + \cdots + 2^{n-6} + 2^{n-5}) = 2^{n-4}$ rows. Subsequently, we iterate through the remaining elements of the set $\mathcal{A}_{n-1}$ in ascending order, starting from its minimal element, xy^{n-2}. For each successive word w, we append the coordinate vector of its derivative $\partial_1(w)$ as a candidate row to $\mathbf{N}_n$; if this candidate row increases the rank of the matrix, it is retained, and otherwise, it is discarded. Upon completion of this procedure, the resulting matrix $\mathbf{N}_n$ is inherently full row-rank, just like the matrix $\mathbf{M}_n$ discussed in Conjecture 1. Furthermore, the ranks of these two matrices are identical.

Since our custom JavaScript application—which implements the aforementioned procedure too—generates a clear graphical layout for small values of n, the visual outputs representing the matrices $\mathbf{N}_6$ and $\mathbf{N}_7$ are included below as illustrative examples.

μ(w)	ρ	D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
w	rank	Delete	xyyyyy	xyyyxy	xyyxyy	xyyxxy	xyxyyy	xyxyxy	xyxxyy	xyxxxy	xxyyyy	xxyyxy	xxyxyy	xxyxxy	xxxyyy	xxxyxy	xxxxyy	xxxxxy
∂_4^1(xy)	1	✕	1	0	1	0	1	0	1	0	0	-1	0	-1	0	-1	0	-1
∂_3^1(xxy)	2	✕	0	1	0	0	0	1	0	0	1	1	1	0	0	0	0	-1
∂_2^1(xxyy)	3	✕	0	0	1	0	0	0	0	0	1	0	0	-1	0	0	-1	0
∂_2^1(xxxy)	4	✕	0	0	0	1	0	0	0	0	0	1	0	1	1	1	0	-1
∂_1^1(xyyyy)	5	✕	1	-1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0	0
∂_1^1(xyyxy)	6	✕	0	1	1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0
∂_1^1(xyxyy)	7	✕	0	0	1	0	1	-1	-1	0	0	0	-1	0	0	0	0	0
∂_1^1(xyxxy)	8	✕	0	0	0	1	0	1	1	-1	0	0	0	-1	0	0	0	0
∂_1^1(xxyyy)	9	✕	0	0	0	0	1	0	0	0	1	-1	-1	0	-1	0	0	0
∂_1^1(xxxyy)	10	✕	0	0	0	0	0	0	1	0	0	0	1	0	1	-1	-1	0

μ(w)	ρ	D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
w	rank	Delete	xyyyyyy	xyyyyyx	xyyyyxy	xyyyyxx	xyyyxyy	xyyyxyx	xyyyxxy	xyyyxxx	xyyxyyy	xyyxyyx	xyyxyxy	xyyxyxx	xyyxxyy	xyyxxyx	xyyxxxy	xyyxxxx	xyxyyyy	xyxyyyx	xyxyyxy	xyxyyxx	xyxyxyy	xyxyxyx	xyxyxxy	xyxyxxx	xyxxyyy	xyxxyyx	xyxxyxy	xyxxyxx	xyxxxyy	xyxxxyx	xyxxxxy	xyxxxxx	xxxxxxx
$\partial_5^1(xy)$	1	✗	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	0	-1	0	-1	0	-1	0	-1	0	-1	0	-1	0	-1	0	-1
$\partial_4^1(xxy)$	2	✗	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	1	1	1	0	1	1	1	0	0	0	0	-1	0	0	0	-1
$\partial_3^1(xxyy)$	3	✗	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	1	0	0	-1	-1	-1	0	0	0	0	0	0	-1	0
$\partial_3^1(xxxy)$	4	✗	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	1	0	1	0	0	1	1	1	1	0	0	0	-1
$\partial_2^1(xxyyy)$	5	✗	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	-1	-1	0	0	-1	0	0	0	0	0	-1	0	0	0
$\partial_2^1(xxyxy)$	6	✗	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	1	0	-1	0	0	0	0	0	-1	0	0
$\partial_2^1(xxxyy)$	7	✗	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	1	0	0	-1	0	0	-1	0
$\partial_2^1(xxxxy)$	8	✗	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	1	0	1	1	1	0	-1
$\partial_1^1(xyyyyyy)$	9	✗	1	-1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
$\partial_1^1(xyyyyxy)$	10	✗	0	1	1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
$\partial_1^1(xyyyxyy)$	11	✗	0	0	1	0	1	-1	-1	0	0	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0	0
$\partial_1^1(xyyyxxy)$	12	✗	0	0	0	1	0	1	1	-1	0	0	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0
$\partial_1^1(xyyxyyy)$	13	✗	0	0	0	0	1	0	0	0	1	-1	-1	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0
$\partial_1^1(xyyxyxy)$	14	✗	0	0	0	0	0	1	0	0	0	1	1	-1	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0
$\partial_1^1(xyyxxyy)$	15	✗	0	0	0	0	0	0	1	0	0	0	1	0	1	-1	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0
$\partial_1^1(xyyxxxy)$	16	✗	0	0	0	0	0	0	0	1	0	0	0	1	0	1	1	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0
$\partial_1^1(xxyyyyy)$	17	✗	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	-1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0	0
$\partial_1^1(xxyyyxy)$	18	✗	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0
$\partial_1^1(xxyyxyy)$	19	✗	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	1	-1	-1	0	0	0	-1	0	0	0	0	0
$\partial_1^1(xxyyxxy)$	20	✗	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	1	1	-1	0	0	0	-1	0	0	0	0
$\partial_1^1(xxxyyyy)$	21	✗	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	1	-1	-1	0	-1	0	0	0
$\partial_1^1(xxxyyxy)$	22	✗	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	1	1	-1	0	-1	0	0

Although it is theoretically possible that a word $w_1 \in \mathcal{A}_{n-1}$ is skipped while the derivative of a larger word $w_2 > w_1$ is retained in the matrix $\mathbf{N}_n$, empirical evidence suggests that this never occurs for $n > 6$. Specifically, for any weight $n \neq 6$, there exists a unique critical threshold word $w_{\text{crit}} \in \mathcal{A}_{n-1}$ such that its derivative is discarded, and consequently, the derivatives of all subsequent words $w > w_{\text{crit}}$ are also eliminated. In short, with the sole exception of $n = 6$, elements are systematically discarded exclusively from the tail of the ordered set $\mathcal{A}_{n-1}$. (The anomalous behavior observed in the case $n = 6$ is explicitly accounted for by the "congestion" of the pentagonal numbers 3 and 5.)

In light of these observations, *Conjecture 2 can be stated in a sharper form as follows*: for any weight $n \neq 6$, during the construction of the matrix $\mathbf{N}_n$, the first and minimal word $w_0 \in \mathcal{A}_{n-1}$ whose derivative fails to increase the rank of the matrix is precisely the $\big(3 \cdot 2^{n-4} - \text{pic}(n-1)\big)$-th element in the ordered set. Consequently, the derivatives of all words larger than w_0 likewise fail to increment the rank of the matrix.

Since $|\mathcal{B}_n^*| = 2^{n-4}$, and exactly $3 \cdot 2^{n-4} - \text{pic}(n-1) - 1$ elements from $\mathcal{A}_{n-1}$ are retained in the matrix $\mathbf{N}_n$, it follows that:

$$\mathscr{L}(n) = \text{rank}(\mathbf{N}_n) = 2^{n-4} + 3 \cdot 2^{n-4} - \text{pic}(n-1) - 1 = 2^{n-2} - \text{pic}(n-1) - 1,$$

which precisely recovers the formulation of Conjecture 2 presented in Section 2. Although the equivalence between Conjecture 1 and Conjecture 2 can be formally established with ease in light of the identities to be discussed in Section 5 (where we shall demonstrate that the rank increments are fundamentally parametric generalizations of pure inverting compositions), we consider it essential to present these two approaches separately, as the structural nuances arising between them are not fully captured by the explicit formulas alone.

It is worth noting that this particular formulation and matrix construction are highly optimized for computationally efficient implementations, which could potentially enable the verification of the conjecture for weights well beyond $n = 17$.

It is highly instructive to reflect upon a comment by N. J. A. Sloane in the OEIS entry for **A178841**, where he noted that the striking similarity between the terms of **A178841**(n) and **A152537**(n) might be "just a coincidence, caused by the fact that their generating functions are almost identical." Our framework provides a rigorous, deeper theoretical explanation that dispels this assumption of coincidence. The connection between these two sequences is fundamentally rooted in the dual matrix constructions presented herein. Specifically, building the derivation matrix in descending order yields a rank-increment structure governed by **A152537** (Conjecture 1), whereas building it in ascending order yields a structure governed by **A178841** (Conjecture 2). Since

both procedures ultimately resolve the exact same system of Ihara-Kaneko-Zagier derivation relations, the ranks of the resulting matrices $\mathbf{M}_n$ and $\mathbf{N}_n$ must identically coincide with $\mathscr{L}(n)$, thereby intrinsically forcing the near-identity of their respective generating functions.

5. Generalization of pic(n)

Preliminary draft.

In this section, we demonstrate that the rank increments can be viewed as a natural generalization of the classical sequence $\mathrm{pic}(k)$ to the parametric function $\mathrm{pic}_n(k)$.

Before presenting the formal identities, we define three fundamental families of polynomials that govern these combinatorial structures. Let Fib_n and Fab_n denote the multivariate Fibonacci and Faber partition polynomials, respectively. Furthermore, let Z_n denote the cycle index polynomial of the symmetric group S_n. For a given set of variables $(x_1, x_2, \ldots, x_n)$, these polynomials are explicitly defined as follows:

$$\begin{aligned}
\mathrm{Fib}_n(x_1, x_2, \ldots x_n) &= \sum_{1b_1+2b_2+\cdots+nb_n=n} (-1)^{b_1+\cdots+b_n}\,(b_1+\cdots+b_n)!\,\frac{x_1^{b_1}x_2^{b_2}\cdots x_n^{b_n}}{b_1!\,b_2!\cdots b_n!}\\
\mathrm{Fab}_n(x_1, x_2, \ldots x_n) &= \sum_{1b_1+2b_2+\cdots+nb_n=n} (-1)^{b_1+\cdots+b_n}\,n\;(b_1+\cdots+b_n-1)!\,\frac{x_1^{b_1}x_2^{b_2}\cdots x_n^{b_n}}{b_1!\,b_2!\cdots b_n!}\\
\mathrm{Z}_n(x_1, x_2, \ldots x_n) &= \sum_{1b_1+2b_2+\cdots+nb_n=n} \frac{1}{1^{b_1}2^{b_2}\cdots n^{b_n}}\,\frac{x_1^{b_1}x_2^{b_2}\cdots x_n^{b_n}}{b_1!\,b_2!\cdots b_n!}
\end{aligned}$$

Motivated by the theoretical results established in [2], we present the fundamental recurrence relation satisfied by $\mathrm{pic}(k)$, along with several formulas that characterize the intricate relations among the pure inverting composition function $\mathrm{pic}(k)$, the partition function $p(k)$, and the divisor sum function $\sigma(k)$.

$$\begin{aligned}
\mathrm{pic}(k) &= \sum_{j=0}^{k} \omega(j)\,2^{k-1-j} + \omega(j)\\
k\,\mathrm{pic}(k) &= \sum_{j=0}^{k} \mathrm{pic}(k-j)\,\left(2^j - \sigma(j) - 1\right)\\
\mathrm{pic}(k) &= 2\,\mathrm{pic}(k-1) + \omega(k) - \omega(k-1)\\
\mathrm{pic}(k) &= \mathrm{Z}_k(2^1-\sigma(1)-1, 2^2-\sigma(2)-1, \ldots, 2^k-\sigma(k)-1)\\
\sigma(k) - 2^k + 1 &= \mathrm{Fab}_k(\mathrm{pic}(1), \mathrm{pic}(2), \ldots, \mathrm{pic}(k))\\
\sum_{j=0}^{k} p(j) - 2p(k) &= -\mathrm{Fib}_k(\mathrm{pic}(1), \mathrm{pic}(2), \ldots, \mathrm{pic}(k))\\
\mathrm{pic}(k) &= \mathrm{Fib}_k\left(2p(1) - \sum_{j=0}^{1} p(j), 2p(2) - \sum_{j=0}^{2} p(j), \ldots, 2p(k) - \sum_{j=0}^{k} p(j)\right)\\
p(k) &= \sum_{j=0}^{k} 2^{\max(0,k-1-j)}\,\mathrm{Fib}_j(\mathrm{pic}(1), \mathrm{pic}(2), \ldots, \mathrm{pic}(j))
\end{aligned}$$

These relationships can be naturally generalized by restricting the underlying arithmetic functions. Specifically, let $p_{>n}(k)$ denote the number of partitions of k into parts strictly greater than n, and let $\sigma_n(k)$ denote the restricted divisor sum function, representing the sum of divisors of k that are at most n. By adopting the first formula below as the formal definition of the generalized function $\mathrm{pic}_n(k)$, all the preceding identities can be seamlessly mapped to this generalized setting as follows:

$$\begin{aligned}
\mathrm{pic}_n(k) &:= \mathrm{Z}_k(2^1-\sigma_n(1)-1, 2^2-\sigma_n(2)-1, \ldots, 2^k-\sigma_n(k)-1)\\
k\,\mathrm{pic}_n(k) &= \sum_{j=1}^{k} \mathrm{pic}_n(k-j)\ \left(2^j-\sigma_n(j)-1\right)\\
\mathrm{pic}_n(k) &= \mathrm{pic}_{n-1}(k)-\mathrm{pic}_{n-1}(k-n)\\
\mathrm{pic}_n\left(\binom{n+1}{2}+k\right) &= 2^{k-1}\left[2\,\mathrm{pic}_n\left(\binom{n+1}{2}\right)-(-1)^n\right]\\
\sigma_n(k)-2^k+1 &= \mathrm{Fab}_k(\mathrm{pic}_n(1), \mathrm{pic}_n(2), \ldots, \mathrm{pic}_n(k))\\
\sum_{j=0}^{k} p_{>n}(j)-2p_{>n}(k) &= -\mathrm{Fib}_k(\mathrm{pic}_n(1), \mathrm{pic}_n(2), \ldots, \mathrm{pic}_n(k))\\
\mathrm{pic}_n(k) &= \mathrm{Fib}_k\left(2p_{>n}(1)-\sum_{j=0}^{1} p_{>n}(j), 2p_{>n}(2)-\sum_{j=0}^{2} p_{>n}(j), \ldots, 2p_{>n}(k)-\sum_{j=0}^{k} p_{>n}(j)\right)\\
p_{>n}(k) &= \sum_{j=0}^{k} 2^{\max(0,k-1-j)}\,\mathrm{Fib}_j(\mathrm{pic}_n(1), \mathrm{pic}_n(2), \ldots, \mathrm{pic}_n(j))
\end{aligned}$$

Furthermore, the following connection holds between the newly introduced generalized function $\mathrm{pic}_n(k)$ and the classical sequence $\mathrm{pic}(k)$:

$$\sum_{k=0}^{n} \mathrm{pic}_{n-k}(k) = 2^n - \mathrm{pic}(n+1)\ (n=0,1,2,\ldots).$$

For all $n \geq 0$ let $\omega_n(k)$ (cf. OEIS:A231599 and A333290) defined by

$$\omega_n(k) := \begin{cases} 1 & \text{if } n,k=0,\\ 0 & \text{if } n=0 \text{ and } k>0,\\ \text{the coefficient of } x^k \text{ in the expansion of the product } \prod_{j=1}^{n}(1-x^j) & \text{if } n>0 \text{ and } k\geq 0 \end{cases}$$

For each $n \geq 0$, we introduce the following sequences:

$$\begin{aligned}
&\omega_n := (\omega_n(0), \omega_n(1), \omega_n(2), \omega_n(3), \ldots), && \mathbf{p} := (p(0), p(1), p(2), p(3), \ldots),\\
&\mathbf{pic} := (\mathrm{pic}(0), \mathrm{pic}(1), \mathrm{pic}(2), \ldots), && \mathbf{p}_{>n} := (p_{>n}(0), p_{>n}(1), p_{>n}(2), \ldots),\\
&\mathbf{pic}_n := (\mathrm{pic}_n(0), \mathrm{pic}_n(1), \mathrm{pic}_n(2), \ldots), && \mathbf{p}_{\leqq n} := (p_{\leqq n}(0), p_{\leqq n}(1), p_{\leqq n}(2), \ldots),
\end{aligned}$$

where $p_{\leqq n}(k)$ denotes the number of partitions of k into parts $\leq n$ (or equivalently, into at most n parts).

($\mathbf{p}_{>n}$: cf. OEIS:A000041, A002865, A008483, A008484, A185325-A185329)

($\mathbf{p}_{\leqq n}$: cf. OEIS:A000012, A004526, A001399, A001400, A001401,A001402, A026813-A026816)

By definition, the following limiting cases immediately follow from the interpretations of these sequences:

$$\omega_0 = (1,0,0,0,\ldots),\ \omega_\infty = \omega,\ \mathbf{p} = \mathbf{p}_{>0} = \mathbf{p}_{\leqq\infty},\ \mathbf{0} = (0,0,0,\ldots) = \mathbf{p}_{>\infty} = \mathbf{p}_{\leqq 0},\ \mathbf{1} = (1,1,1,\ldots) = \mathbf{p}_{\leqq 1}.$$

We will use the well-known inverse relation

$$\omega_n^{-*} = \mathbf{p}_{\leqq n}\ (n = 0,1,2,\ldots,\infty)$$

where the superscript $\mathbf{a}^{-*}$ denotes the convolutional inverse of a sequence $\mathbf{a}$.

Using these algebraic formulations and sequence definitions, the relations established above for $\mathrm{pic}_n(k)$ can be rewritten in a highly compact, equivalent form via the following convolutions:

$$\begin{aligned}
\mathbf{p}_{>n} &= \omega_n * \mathbf{p}\\
\mathbf{pic}_n &= \mathbf{p}_{>n} * \mathbf{pic}
\end{aligned}$$

To illustrate these algebraic properties, we explicitly compute the initial terms of the sequences $\mathbf{pic}_n$ corresponding to the cases $n = 0, 1,$ and 2.

$$\begin{aligned}
\mathbf{pic}_0 &= \mathbf{p}_{>0} * \mathbf{pic} = \mathbf{p} * \mathbf{pic} \\
&= (1,1,2,3,5,7,11,15,22,30,42,56,77,101,\ldots) * (1,0,0,1,2,5,9,19,37,74,148,296,591,1183,\ldots) \\
&= (1,1,2,4,8,16,32,64,128,256,512,1024,2048,4096,\ldots). \\
\mathbf{pic}_1 &= \mathbf{p}_{>1} * \mathbf{pic} \\
&= (1,0,1,1,2,2,4,4,7,8,12,14,21,24,\ldots) * (1,0,0,1,2,5,9,19,37,74,148,296,591,1183,\ldots) \\
&= (1,0,1,2,4,8,16,32,64,128,256,512,1024,2048,\ldots). \\
\mathbf{pic}_2 &= \mathbf{p}_{>2} * \mathbf{pic} \\
&= (1,0,0,1,1,1,2,2,3,4,5,6,9,10,\ldots) * (1,0,0,1,2,5,9,19,37,74,148,296,591,1183,\ldots) \\
&= (1,0,0,2,3,6,12,24,48,96,192,384,768,1536,\ldots).
\end{aligned}$$

The following matrix summarizes the first 14 terms of the sequences $\mathbf{pic}_n$ computed for $n = 0, 1, \ldots, 9$.

$$\begin{bmatrix}
1 & 1 & 2 & 4 & 8 & 16 & 32 & 64 & 128 & 256 & 512 & 1024 & 2048 & 4096 \\
1 & 0 & 1 & 2 & 4 & 8 & 16 & 32 & 64 & 128 & 256 & 512 & 1024 & 2048 \\
1 & 0 & 0 & 2 & 3 & 6 & 12 & 24 & 48 & 96 & 192 & 384 & 768 & 1536 \\
1 & 0 & 0 & 1 & 3 & 6 & 10 & 21 & 42 & 84 & 168 & 336 & 672 & 1344 \\
1 & 0 & 0 & 1 & 2 & 6 & 10 & 20 & 39 & 78 & 158 & 315 & 630 & 1260 \\
1 & 0 & 0 & 1 & 2 & 5 & 10 & 20 & 38 & 76 & 152 & 305 & 610 & 1221 \\
1 & 0 & 0 & 1 & 2 & 5 & 9 & 20 & 38 & 75 & 150 & 300 & 600 & 1201 \\
1 & 0 & 0 & 1 & 2 & 5 & 9 & 19 & 38 & 75 & 149 & 298 & 595 & 1192 \\
1 & 0 & 0 & 1 & 2 & 5 & 9 & 19 & 37 & 75 & 149 & 297 & 593 & 1187 \\
1 & 0 & 0 & 1 & 2 & 5 & 9 & 19 & 37 & 74 & 149 & 297 & 592 & 1185 \\
1 & 0 & 0 & 1 & 2 & 5 & 9 & 19 & 37 & 74 & 148 & 297 & 592 & 1184
\end{bmatrix}$$

By shifting each row of this matrix by its corresponding index $n = 0, 1, 2, 3, \ldots$, we obtain precisely the matrix $\mathbf{R}$ discussed in Conjecture 1:

$$\mathbf{R} = \begin{bmatrix}
1 & 1 & 2 & 4 & 8 & 16 & 32 & 64 & 128 & 256 & 512 & 1024 & 2048 & 4096 & \cdots \\
0 & 1 & 0 & 1 & 2 & 4 & 8 & 16 & 32 & 64 & 128 & 256 & 512 & 1024 & \cdots \\
0 & 0 & 1 & 0 & 0 & 2 & 3 & 6 & 12 & 24 & 48 & 96 & 192 & 384 & \cdots \\
0 & 0 & 0 & 1 & 0 & 0 & 1 & 3 & 6 & 10 & 21 & 42 & 84 & 168 & \cdots \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 2 & 6 & 10 & 20 & 39 & 78 & \cdots \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 2 & 5 & 10 & 20 & 38 & \cdots \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 2 & 5 & 9 & 20 & \cdots \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 2 & 5 & 9 & \cdots \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 2 & 5 & \cdots \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 2 & \cdots \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & \cdots
\end{bmatrix}$$

By combining the two fundamental convolution formulas above, we immediately obtain the following identity for $\mathbf{pic}_n$:

$$\mathbf{pic}_n = \mathbf{p}_{>n} * \mathbf{pic} = \omega_n * (\mathbf{p} * \mathbf{pic}) = \omega_n * (1,1,2,4,8,16,32,\ldots)$$

Convolving both sides of this equation with ω_n^{-*}, we arrive at:

$$\omega_n^{-*} * \mathbf{pic}_n = \mathbf{p}_{\leq n} * \mathbf{pic}_n = \mathbf{p} * \mathbf{pic} = (1,1,2,4,8,16,32,\ldots)$$

Alternatively, convolving both sides with $\mathbf{p}^{-*} = \omega$ we get the remarkably symmetric relation:

$$\omega * \mathbf{pic}_n = \omega_n * \mathbf{pic}$$

6. The Pentagonal Transform and Its Recurrence Matrix

Let S denote the left shift operator acting on sequences, defined by

$$S((a_n)_{n\geq 0}) = (a_{n+1})_{n\geq 0}.$$

For each $k \geq 0$, let $\mathbf{e}_k = (e_k(n))_{n\geq 0}$ denote the k-th standard basis sequence, defined via the Kronecker delta by $e_k(n) := \delta_{k,n} \quad (n \geq 0)$. In particular, $\mathbf{e}_0 = (1, 0, 0, \dots)$ and $\mathbf{e}_1 = (0, 1, 0, \dots)$, where the only non-zero term of $\mathbf{e}_k$ appears at index k. We will frequently use throughout this chapter the sequence $\delta = (\delta(n))_{n\geq 0}$

$$\delta := S(\mathbf{e}_0 - \omega) = (1, 1, 0, 0, -1, 0, -1, 0, 0, 0, 0, 1, 0, 0, 1, 0, \dots).$$

That is, $\delta(n)$ =A257628(n+1) $(n \geq 0)$. We retain the notation for the vectors ω_n, $\mathbf{p}$, $\mathbf{p}_{\leqq n}$, $\mathbf{p}_{>n}$, $\mathbf{pic}$, $\mathbf{pic}_n$ established in the previous chapter.

Definition. Let $\mathbf{a} = (a(n))_{n\geq 0}$ be a fixed sequence with $a(0) = 1$. We associate with $\mathbf{a}$ an infinite two-dimensional matrix $\mathbf{R}(\mathbf{a}) = (r(i, j))$ $(i, j \in \mathbb{Z})$ by the following recurrence relation:

$$r(i,j) = \begin{cases} 0 & \text{if } i < 0 \text{ or } j < i, \\ a(j) & \text{if } i = 0 \text{ and } j \geq 0, \\ r(i-1, j-1) - r(i-1, j-1-i) & \text{otherwise.} \end{cases}$$

The *pentagonal transform* of the sequence $\mathbf{a}$, denoted by $\mathcal{P}(\mathbf{a})$ is the sequence $\mathbf{b} = (b(n))_{n\geq 0}$ whose n-th term is given by the sum of the elements in the n-th column of $\mathbf{R}(\mathbf{a})$:

$$b(n) := \sum_{i=0}^{n} r(i, n) \quad (n \geq 0).$$

It is an immediate consequence of the definition that the transform $\mathcal{P}$ is linear. The pentagonal transform $\mathcal{P}$ and its underlying recurrence relation possess widespread applications; in fact, the wealth of combinatorial identities derivable from them could fill an entire volume on their own. For our present purposes, however, it suffices to note that for any arbitrary sequence $\mathbf{a}$, we have

$$\mathcal{P}(\mathbf{a}) = \delta * \mathbf{a},$$

where $*$ denotes the convolution of sequences.

To illustrate the mechanics of the pentagonal transform $\mathcal{P}$ and the structure of its associated matrix $\mathbf{R}(\mathbf{a})$, we present two concrete examples generated via a custom JavaScript application. Since these matrices are inherently infinite, the figures display only their first few rows and columns. The first displays matrix $\mathbf{R}(\mathbf{e}_0)$ corresponding to the standard basis sequence $\mathbf{e}_0 = (1, 0, 0, \dots)$ (indicated in the top blue row). As expected, the resulting column sums (highlighted in the bottom green row) precisely yield the sequence δ, reflecting the property $\mathcal{P}(\mathbf{e}_0) = \delta$. Similarly, the second demonstrates the transform applied to the sequence of partition numbers $\mathbf{p} = (p(n))_{n\geq 0}$. In this case, the top blue row contains the initial partition values, while the bottom green row displays the row sums constituting the transformed sequence $\mathcal{P}(\mathbf{p}) = S(\mathbf{p})$.

0	1	0	0	0	0	0	0	0	0	0	0	0
1	0	1	-1	0	0	0	0	0	0	0	0	0
2	0	0	1	-1	-1	1	0	0	0	0	0	0
3	0	0	0	1	-1	-1	0	1	1	-1	0	0
4	0	0	0	0	1	-1	-1	0	0	2	0	0
5	0	0	0	0	0	1	-1	-1	0	0	1	1
6	0	0	0	0	0	0	1	-1	-1	0	0	1
7	0	0	0	0	0	0	0	1	-1	-1	0	0
8	0	0	0	0	0	0	0	0	1	-1	-1	0
9	0	0	0	0	0	0	0	0	0	1	-1	-1
10	0	0	0	0	0	0	0	0	0	0	1	-1
11	0	0	0	0	0	0	0	0	0	0	0	1
12	0	0	0	0	0	0	0	0	0	0	0	0
Σ	1	1	0	0	-1	0	-1	0	0	0	0	1

0	1	1	2	3	5	7	11	15	22	30	42	56
1	0	1	0	1	1	2	2	4	4	7	8	12
2	0	0	1	0	0	1	1	1	2	2	3	4
3	0	0	0	1	0	0	0	1	1	1	1	2
4	0	0	0	0	1	0	0	0	0	1	1	1
5	0	0	0	0	0	1	0	0	0	0	0	1
6	0	0	0	0	0	0	1	0	0	0	0	0
7	0	0	0	0	0	0	0	1	0	0	0	0
8	0	0	0	0	0	0	0	0	1	0	0	0
9	0	0	0	0	0	0	0	0	0	1	0	0
10	0	0	0	0	0	0	0	0	0	0	1	0
11	0	0	0	0	0	0	0	0	0	0	0	1
12	0	0	0	0	0	0	0	0	0	0	0	0
Σ	1	2	3	5	7	11	15	22	30	42	56	77

Furthermore, the next figure illustrates the transform applied to the sequence $\mathbf{a} = \mathbf{p} * \mathbf{pic} = (1, 1, 2, 4, 8, 16, \dots)$. This particular case is of primary interest as it is directly related to Conjecture 1, by virtue of the fact that each value of the sequence $\mathcal{P}(\mathbf{a})$ displayed in the bottom green row is precisely one greater than the corresponding value of $\mathcal{L}(n)$.

0	1	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536	131072	262144
1	0	1	0	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536
2	0	0	1	0	0	2	3	6	12	24	48	96	192	384	768	1536	3072	6144	12288	24576
3	0	0	0	1	0	0	1	3	6	10	21	42	84	168	336	672	1344	2688	5376	10752
4	0	0	0	0	1	0	0	1	2	6	10	20	39	78	158	315	630	1260	2520	5040
5	0	0	0	0	0	1	0	0	1	2	5	10	20	38	76	152	305	610	1221	2442
6	0	0	0	0	0	0	1	0	0	1	2	5	9	20	38	75	150	300	600	1201
7	0	0	0	0	0	0	0	1	0	0	1	2	5	9	19	38	75	149	298	595
8	0	0	0	0	0	0	0	0	1	0	0	1	2	5	9	19	37	75	149	297
9	0	0	0	0	0	0	0	0	0	1	0	0	1	2	5	9	19	37	74	149
10	0	0	0	0	0	0	0	0	0	0	1	0	0	1	2	5	9	19	37	74
11	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	2	5	9	19	37
12	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	2	5	9	19
13	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	2	5	9
14	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	2	5
15	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	2
16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1
17	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
19	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
∑	1	2	3	6	11	23	45	91	182	364	728	1457	2913	5826	11653	23305	46610	93220	186440	372880

If $\mathbf{1}$ denotes the constant sequence $(1, 1, 1, \dots)$ and $\mathbf{L}$ denotes the sequence $(\mathcal{L}(2), \mathcal{L}(3), \mathcal{L}(4), \dots)$, then this relationship can be expressed in the following compact form

$$\mathbf{L} = \mathcal{P}(\mathbf{a}) - \mathbf{1} = \delta * \mathbf{p} * \mathbf{pic} - \mathbf{1},$$

from which the formula stated in Conjecture 1 can be obtained via basic algebraic manipulations.

However, the connection runs much deeper; it manifests not only at the level of the transformed sequence $\mathbf{L} = \mathcal{P}(\mathbf{a}) - \mathbf{1}$, but also throughout the entire recurrence matrix, appearing in every single row. Consequently, this observation opens a clear avenue toward proving Conjecture 1. To elucidate this deeper connection, we introduce a so-called rank-increment matrix associated with the derivation relation, and compare its empirically observed values with the matrix of the recently discussed transform. In alignment with the notation established in Chapter 3, let $\mathcal{A}_n := x\mathfrak{H}_{n-2}y$ denote the set of admissible words of length $n \geq 2$, and let $\partial_m(\mathcal{A}_n) := \{\partial_m(w) : w \in \mathcal{A}_n\}$ for any $m \geq 1$. For notational convenience, we index the rows of the rank-increment matrix $\mathbf{R} = (r_{m,n})$ starting from 1 and its columns starting from 2, so that $m \geq 1$ and $n \geq 2$. The entries $r_{n,m}$ of the rank-increment matrix are formally defined as follows:

(1) The entry in the first row and n-th column is given by the rank of the vector system $\partial_1(\mathcal{A}_{n-1})$; that is,

$$r_{1,n} = \operatorname{rank}\left(\partial_1(\mathcal{A}_{n-1})\right) \quad \text{for all } n \geq 2.$$

(2) For the subsequent rows where $m \geq 2$, the entry $r_{m,n}$ is defined as the rank increment obtained by adjoining the vector system $\partial_m(\mathcal{A}_{n-m})$ to the combined system

$$\bigcup_{k=1}^{m-1} \partial_k(\mathcal{A}_{n-k}).$$

That is,

$$r_{m,n} = \operatorname{rank}\left(\bigcup_{k=1}^{m} \partial_k(\mathcal{A}_{n-k})\right) - \operatorname{rank}\left(\bigcup_{k=1}^{m-1} \partial_k(\mathcal{A}_{n-k})\right)$$

for all $m \geq 2$ and $n \geq m + 1$.

This formulation is completely equivalent to the rank-increment procedure discussed in Chapter 3; however, while the focus here is on conceptual clarity and conciseness, the description there was tailored to precisely reflect the algorithmic implementation. By exploiting the underlying duality, for rows where $2 \leq m \leq n - 3$, we restricted our analysis to the vector systems $\partial_m(x\mathcal{A}_{n-m-1})$, which comprise only half of the full system $\partial_m(\mathcal{A}_{n-m})$. Furthermore, those elements were adjoined to the matrix sequentially, one by one, according to a fixed ordering, discarding those that did not increase the rank.

The empirically computed rank-increment matrix is presented below, exhibiting a perfect match with the strictly upper triangular part of the transformation matrix $\mathbf{R}(\mathbf{a})$ initialized with the sequence $\mathbf{a} = (1, 1, 2, 4, 8, 16, \dots)$.

	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
∂_1	1	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384
∂_2			0	1	2	4	8	16	32	64	128	256	512	1024	2048	4096
∂_3				0	0	2	3	6	12	24	48	96	192	384	768	1536
∂_4					0	0	1	3	6	10	21	42	84	168	336	672
∂_5						0	0	1	2	6	10	20	39	78	158	315
∂_6							0	0	1	2	5	10	20	38	76	152
∂_7								0	0	1	2	5	9	20	38	75
∂_8									0	0	1	2	5	9	19	38
∂_9										0	0	1	2	5	9	19
∂_{10}											0	0	1	2	5	9
∂_{11}												0	0	1	2	5
∂_{12}													0	0	1	2
∂_{13}														0	0	1
∂_{14}															0	0
∂_{15}																0
∂_{16}																

The fact that the powers of two appear in the first row of the empirical rank-increment matrix $\mathbf{R}$ is theoretically straightforward to verify; specifically, $\operatorname{rank}(\partial_1(\mathcal{A}_n)) = 2^{n-3}$ for $n \geq 3$. Indeed, the cardinality of the underlying set is given by $|\mathcal{A}_{n-1}| = |x\mathfrak{H}_{n-3}y| = 2^{n-3}$. To establish the linear independence of the vector system $\partial_1(\mathcal{A}_{n-1})$, it suffices to examine the first term of the derivative of each word $w \in \mathcal{A}_n$ with respect to the ordering fixed in Chapter 3. Consequently, to complete the proof of Conjecture 1, it suffices to demonstrate that under the appropriate reindexing, the defined rank increments $r_{m,n}$ satisfy the recurrence relation prescribed in the definition of the pentagonal transform.

To further investigate the underlying algebraic foundations, it is highly instructive that the matrix can be viewed as a two-dimensional extension $\mathbf{pic}_n(k)$ of the sequence of pure inverting compositions $\mathbf{pic}(k)$, as discussed in detail in the preceding section. In view of what is described there, *Conjecture 1 would follow from directly showing that* the vector system $\partial_m(\mathcal{A}_k)$ increases the rank of the system

$$\bigcup_{j=1}^{m-1} \partial_{m-j}(\mathcal{A}_{k+j}) = \partial_{m-1}(\mathcal{A}_{k+1}) \cup \partial_{m-2}(\mathcal{A}_{k+2}) \cup \cdots \cup \partial_1(\mathcal{A}_{k+m-1})$$

by exactly

$$\sum_{i=0}^{k} p_{>m}(i) \cdot \operatorname{pic}(k-i) = p_{>m}(0)\operatorname{pic}(k) + p_{>m}(1)\operatorname{pic}(k-1) + \cdots + p_{>m}(k-1)\operatorname{pic}(1) + p_{>m}(k)\operatorname{pic}(0).$$

To conclude this chapter, we provide an algebraic formulation of the pentagonal transform using the framework of generating functions. Let $A(x) = \sum_{k=0}^{\infty} a(k)x^k$ be the ordinary generating function of an arbitrary sequence $\mathbf{a}$ with $a(0) = 1$. Specifically, the generating function for the k-th row of the transformation matrix is given by

$$A(x)\, x^k \prod_{j=1}^{k}(1-x^j) \quad (k = 0, 1, 2, \ldots).$$

Consequently, the generating function of the pentagonal transform of the sequence $\mathbf{a}$ can be expressed as:

$$\mathcal{P}(A)(x) = \sum_{k=0}^{\infty} A(x)\, x^k \prod_{j=1}^{k}(1-x^j) = A(x) \sum_{k=0}^{\infty} x^k \prod_{j=1}^{k}(1-x^j).$$

By applying Euler's Telescoping Lemma, the latter summation can be simplified directly to the form

$$\sum_{k=1}^{\infty} \delta(k)x^k,$$

thus

$$\mathcal{P}(A)(x) = A(x) \sum_{k=1}^{\infty} \delta(k)x^k = \sum_{j=0}^{\infty} x^j \sum_{l=0}^{j-1} a(l)\delta(j-l)$$

which successfully recovers the global generating function of the pentagonal sequence δ and confirms the validity of the convolution property.

7. Other Conjectures on Derivation Relations

Recall that for any admissible word $w \in \mathfrak{H}_0$ and any integers $a, b \geq 0$, the shuffle regularization of the word $y^a w x^b$ is given by

$$\mathrm{reg}_{ш}(y^a w x^b) := \sum_{i=0}^{a} \sum_{j=0}^{b} (-1)^{i+j}\, y^i \,ш\, \left(y^{a-i} w x^{b-j}\right) ш\, x^j$$

Conjecture 4. *(Generalized Derivation Relation) For any admissible word $w \in \mathfrak{H}_0$ and any integers $n > 0$, $a, b \geq 0$, and $m > a + b$, we have*

$$\zeta\left(reg_{ш}\left[\partial_n^m\left(y^a w x^b\right)\right]\right) = 0.$$

Moreover, the quasi-derivation relation can be generalized in a similar manner:

Conjecture 5. *(Generalized Quasi-derivation Relation) For any rational number $c \in \mathbb{Q}$, any admissible word $w \in \mathfrak{H}_0$ and any integers $n > 0$, $a, b \geq 0$, and $m > a + b$, we have*

$$\zeta\left(reg_{ш}\left[\left(\partial_n^{(c)}\right)^m\left(y^a w x^b\right)\right]\right) = 0.$$

The bold rows of the table (which also incorporate rows from the table in [5]) below present the number of linearly independent relations obtained from Conjecture 4, Conjecture 5 (calculated via a JavaScript application developed by the author), and the linear part of Kawashima's relation, respectively.

Table 5. Number of Independent Relations for MZVs

n	3	4	5	6	7	8	9	10	11	12	13	14	15
∂_n	1	2	5	10	22	44	90	181	363	727	1456	2912	5825
∂_n^m	**1**	**2**	**5**	**10**	**24**	**50**	**104**	**213**	**431**	**873**	**1756**	**3532**	**7080**
$\partial_n^{(c)}$	1	2	5	10	23	46	98	200	410	830	1679	-	-
$\left(\partial_n^{(c)}\right)^m$	**1**	**2**	**5**	**11**	**24**	**50**	**105**	**217**	**445**	**910**	-	-	-
Ohno	1	2	5	10	23	46	98	199	411	830	1691	-	-
Lin.Kaw.	**1**	**2**	**5**	**10**	**23**	**46**	**98**	**200**	**413**	**838**	**1713**	-	-
All	1	3	6	14	29	60	123	249	503	1012	2032	4075	8164

8. Conjecture 15 in [2]

A proposed proof.

We will use both the function $\omega(k)$ (cf. OEIS A010815(k)) introduced in Section 1 and the function $\omega_n(k)$ (cf. OEIS A231599(n,k)) defined in Section 5. The latter satisfies

$$\omega_n(k) = \mathrm{Z}_k(-\sigma_n(1), -\sigma_n(2), \ldots, -\sigma_n(k)) = \left[x^k\right] \prod_{j=1}^{n} (1 - x^j),$$

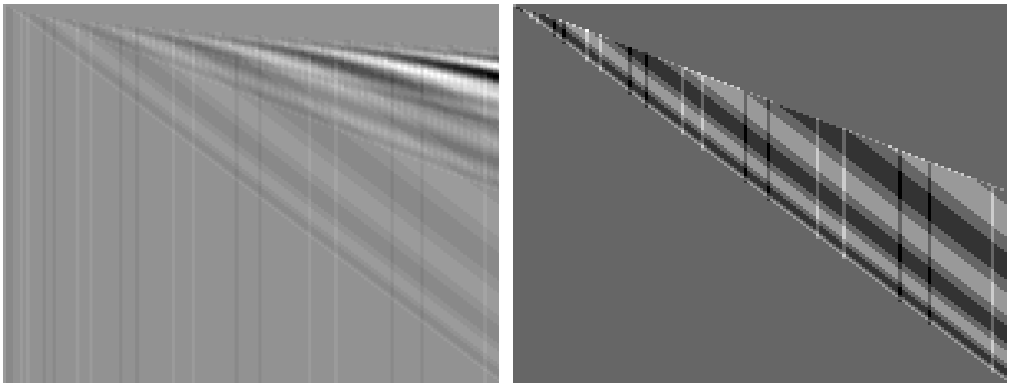
while

$$\omega(k) = \omega_\infty(k) = \mathrm{Z}_k(-\sigma(1), -\sigma(2), \ldots, -\sigma(k)) = \left[x^k\right] \prod_{j=1}^{\infty} (1 - x^j),$$

where $\sigma_n(k)$ denotes the sum of divisors of k that are at most n. Moreover, the values $\omega_n(k)$ satisfy the recurrence relation

$$\omega_n(k) = \omega_{n-1}(k) - \omega_{n-1}(k - n).$$

The image below visualizes the values of the matrix $\Omega_{n,k} = \omega_n(k)$ for $0 \leq n, k \leq 150$, where black corresponds to -16 and white to 12. The structure of triangular regions originating from the origin is clearly visible. Within each such region, the bands run parallel to the lower boundary of the domain. (The vertical lines descend at the generalized pentagonal numbers, that is, at those values of k where $\omega(k) \neq 0$.) *Essentially, the conjecture concerns the prominent bands running parallel to the main diagonal in the region $n \leq k \leq 2n + 1$.* This phenomenon is precisely explained by the statement below, which is the central claim of the entire proof.



Claim 6. Let n be an arbitrary non-negative integer. Then, for every integer k satisfying $n \le k \le 2n+1$, the following relation holds:

$$\omega_n(k) = \omega(k) + \sum_{i=0}^{k-n-1} \omega(i).$$

Proof. Let $\Omega(x) = \prod_{j=1}^{\infty}(1-x^j) = \sum_{j=0}^{\infty}\omega(j)x^j$ and $\Omega_n(x) = \prod_{j=1}^{n}(1-x^j) = \sum_{j=0}^{\infty}\omega_n(j)x^j$. By definition, we have

$$\Omega_n(x) = \Omega(x)\left(\prod_{j=n+1}^{\infty}(1-x^j)\right)^{-1}.$$

In the infinite product $\prod_{j=n+1}^{\infty}(1-x^j) = (1-x^{n+1})(1-x^{n+2})\cdots(1-x^{2n+1})(1-x^{2n+2})\cdots$, multiplying any two or more non-constant components produces a term of degree at least $2n+3$. Thus, any contribution to degrees strictly less than $2n+2$ must come from choosing the non-constant term from at most one factor. Therefore,

$$\prod_{j=n+1}^{\infty}(1-x^j) = 1 - x^{n+1} - x^{n+2} - \cdots - x^{2n+1} - \mathcal{O}(x^{2n+2}).$$

Consequently, to determine the coefficients up to order $2n+1$ in the corresponding power series expansion, it suffices to examine the truncated inverse $\left(1-\sum_{i=n+1}^{2n+1}x^i\right)^{-1}$. Expanding this via a standard geometric series gives

$$\left(1-\sum_{i=n+1}^{2n+1}x^i\right)^{-1} = 1 + \sum_{i=n+1}^{2n+1}x^i + \left(\sum_{i=n+1}^{2n+1}x^i\right)^2 + \cdots.$$

Since the lowest degree of $\left(\sum_{i=n+1}^{2n+1}x^i\right)^m$ for $m \ge 2$ is $2(n+1) = 2n+2$, we obtain

$$\left(\prod_{j=n+1}^{\infty}(1-x^j)\right)^{-1} = 1 + \sum_{i=n+1}^{2n+1}x^i + \mathcal{O}(x^{2n+2}),$$

and thus

$$\Omega_n(x) = \left(\sum_{j=0}^{\infty}\omega(j)x^j\right)\left(1 + \sum_{j=n+1}^{2n+1}x^j + \mathcal{O}(x^{2n+2})\right).$$

In this product, the coefficient of x^k for $n \le k \le 2n+1$ is $\omega(k) + \sum_{j=n+1}^{k}\omega(k-j)$. Substituting $i = k-j$ into the summation transforms the index range to $0 \le i \le k-n-1$, which gives $\sum_{i=0}^{k-n-1}\omega(i)$. This proves that for any $n \le k \le 2n+1$,

$$\omega_n(k) := \left[x^k\right]\Omega_n(x) = \omega(k) + \sum_{i=0}^{k-n-1}\omega(i),$$

which completes the proof. □

Corollary 7. *For any non-negative integer j, we have*

$$\omega_j(j+1) = \omega(j+1) + 1.$$

Proof. Apply the previous *Claim* 6 by setting $n = j$ and $k = j + 1$. Since the condition $j \le j + 1 \le 2j + 1$ holds for all $j \ge 0$, the identity implies

$$\omega_j(j+1) = \omega(j+1) + \sum_{i=0}^{0} \omega(i) = \omega(j+1) + \omega(0).$$

The result follows immediately since $\omega(0) = 1$ by definition. □

Corollary 8. *For any non-negative integer j, the following identity holds:*

$$\omega_j(2j+2) = \omega(2j+2) + \sum_{l=0}^{j+1} \omega(l) + 1.$$

Proof. Setting $n = j + 1$ and $k = 2j + 2$ in *Claim* 6 satisfies the condition $n \le k \le 2n + 1$. Direct substitution yields the evaluation for the higher index:

$$\omega_{j+1}(2j+2) = \omega(2j+2) + \sum_{l=0}^{j} \omega(l).$$

By the recurrence relation $\omega_n(k) = \omega_{n-1}(k) - \omega_{n-1}(k-n)$ with $n = j + 1$ and $k = 2j + 2$, we express the left-hand side as

$$\omega_{j+1}(2j+2) = \omega_j(2j+2) - \omega_j(j+1).$$

Rearranging terms and applying *Corollary* 7 to substitute $\omega_j(j+1) = \omega(j+1) + 1$ leads to

$$\omega_j(2j+2) = \omega_{j+1}(2j+2) + \omega(j+1) + 1.$$

Combining these relations and incorporating $\omega(j+1)$ into the summation concludes the proof. □

Our goal is to show that for every positive integer i, the function $P_i(x) := \left(S_{-i}(x)x^i - S_0(x)C(x)^{-i}\right)(1-x)^i$ is polynomial of degree $3i - 2$. Our entirely elementary and constructive approach provides the exact formula for the polynomials $P_i(x)$.

To this end, we first establish the fundamental relations underpinning our computations and clarify their connection to the notation and preliminaries introduced in [2]. Using the function ω defined above, the Knuth–Netto formula from [2] can be written in the following form: for arbitrary non-negative integers n and $k \le n$, we have

$$I_n(k) = \sum_{m=0}^{k} \binom{n+k-1-m}{n-1} \omega(m). \tag{K-N}$$

We shall also rely on a generalization of the Knuth–Netto formula that holds for arbitrary non-negative integers n and k, without the restriction $k \le n$:

$$I_n(k) = \sum_{m=0}^{k} \binom{n+k-1-m}{n-1} \omega_n(m). \tag{g-K-N}$$

Setting $k = n - i$ in (g-K-N) provides

$$I_n(n-i) = \sum_{m=0}^{n-i} \binom{2n-i-1-m}{n-1} \omega_n(m),$$

which allows us to represent $S_i(x)$ for any integer $i \in \mathbb{Z}$ as

$$S_i(x) := \sum_{n=0}^{\infty} I_n(n-i)x^n = \sum_{n=0}^{\infty} x^n \sum_{m=0}^{n-i} \binom{2n-i-1-m}{n-1} \omega_n(m).$$

Adopting the notation established in [2], this series can be rewritten for $i \ge 0$ as

$$S_{-i}(x) = \sum_{n=0}^{\infty} x^n \sum_{m=0}^{n+i} \binom{2n+i-1-m}{n-1} \omega_n(m).$$

In particular, the case $i = 0$ reduces to

$$S_0(x) = \sum_{n=0}^{\infty} I_n(n)x^n = \sum_{n=0}^{\infty} x^n \sum_{m=0}^{n} \binom{2n-1-m}{n-1} \omega_n(m).$$

Finally, it is a well-known that the generating function for $C(x)^{-i}$ can be expressed as:

$$C(x)^{-i} = 1 - i \sum_{n=1}^{\infty} \frac{1}{n} \binom{2n-i-1}{n-1} x^n$$

We now proceed to compute the difference $S_{-i}(x)x^i - S_0(x)C(x)^{-i}$ for a fixed *positive* integer i. First, let us calculate the power series of the function $S_0(x)C(x)^{-i}$:

$$S_0(x)C(x)^{-i} = \left(1+\sum_{n=1}^{\infty} I_n(n)x^n\right)\left(1 - i\sum_{n=1}^{\infty}\frac{1}{n}\binom{2n-i-1}{n-1}x^n\right) =$$

$$= 1 + \sum_{n=1}^{\infty} I_n(n)x^n - i\sum_{n=1}^{\infty}\frac{1}{n}\binom{2n-i-1}{n-1}x^n - i\left(\sum_{n=1}^{\infty} I_n(n)x^n\right)\left(\sum_{n=1}^{\infty}\frac{1}{n}\binom{2n-i-1}{n-1}x^n\right) =$$

$$= 1 + \sum_{n=1}^{\infty}\left[I_n(n) - \frac{i}{n}\binom{2n-i-1}{n-1}\right]x^n - i\sum_{n=2}^{\infty} x^n \sum_{k=1}^{n-1}\frac{1}{k}\binom{2k-i-1}{k-1} I_{n-k}(n-k) =$$

$$= 1 - ix + \sum_{n=2}^{\infty}\left[I_n(n) - \frac{i}{n}\binom{2n-i-1}{n-1} - i\sum_{k=1}^{n-1}\frac{1}{k}\binom{2k-i-1}{k-1} I_{n-k}(n-k)\right]x^n.$$

Next, we also compute the power series of $S_{-i}(x)x^i$:

$$S_{-i}(x)x^i = \sum_{N=0}^{\infty} x^{N+i}\sum_{m=0}^{N+i}\binom{2N+i-1-m}{N-1}\omega_N(m) = \sum_{n=i}^{\infty} x^n \sum_{m=0}^{n}\binom{2n-i-1-m}{n-i-1}\omega_{n-i}(m),$$

where the second sum is obtained from the first by the change of index $N+i=n$. Thus, for the difference of the two functions, we can write:

$S_{-i}(x)x^i - S_0(x)C(x)^{-i} =$

$$= \sum_{n=i}^{\infty} x^n \sum_{m=0}^{n}\binom{2n-i-1-m}{n-i-1}\omega_{n-i}(m) - 1 + ix - \sum_{n=2}^{\infty}\left[I_n(n) - \frac{i}{n}\binom{2n-i-1}{n-1} - i\sum_{k=1}^{n-1}\frac{1}{k}\binom{2k-i-1}{k-1} I_{n-k}(n-k)\right]x^n =$$

$$= -1 + ix - \sum_{n=2}^{i-1}\left[I_n(n) - \frac{i}{n}\binom{2n-i-1}{n-1} - i\sum_{k=1}^{n-1}\frac{1}{k}\binom{2k-i-1}{k-1} I_{n-k}(n-k)\right]x^n +$$

$$+ \sum_{n=i}^{\infty} x^n\left(\sum_{m=0}^{n}\binom{2n-i-1-m}{n-i-1}\omega_{n-i}(m) - I_n(n) + \frac{i}{n}\binom{2n-i-1}{n-1} + i\sum_{k=1}^{n-1}\frac{1}{k}\binom{2k-i-1}{k-1} I_{n-k}(n-k)\right).$$

This leads to the decomposition $S_{-i}(x)x^i - S_0(x)C(x)^{-i} := A_i(x) + B_i(x)$, where

$$\begin{aligned} A_i(x) &= -1 + i\,x - \sum_{n=2}^{i-1}\left(I_n(n) - \frac{i}{n}\binom{2n-i-1}{n-1} - i\sum_{k=1}^{n-1}\frac{1}{k}\binom{2k-i-1}{k-1} I_{n-k}(n-k)\right)x^n \\ B_i(x) &= \sum_{n=i}^{\infty}\left(\sum_{k=0}^{n}\binom{2n-i-1-k}{n-i-1}\omega_{n-i}(k) - I_n(n) + \frac{i}{n}\binom{2n-i-1}{n-1} + i\sum_{k=1}^{n-1}\frac{1}{k}\binom{2k-i-1}{k-1} I_{n-k}(n-k)\right)x^n \end{aligned}$$

Since the degree of $A_i(x)$ is at most $i-1$, implying that $\deg\left(P_{i,A}(x)\right) := \deg\left(A_i(x)(1-x)^i\right) \le 2i-1$, it suffices to show that $P_{i,B}(x) := B_i(x)(1-x)^i$ is also a polynomial with degree $3i-2$. (The polynomials $P_{i,A}(x)$, $P_{i,B}(x)$, and $P_i(x)$ for values of i from 2 to 6 are presented in Table 8.) Let

$$B_i(x) := \sum_{n=i}^{\infty} b_i(n)x^n \quad \text{and} \quad B_i(x)(1-x)^i := \sum_{n=i}^{\infty} T_i(n)x^n.$$

(C) In terms of these coefficients, the conjecture is equivalent to stating that $T_i(3i-2) \neq 0$ and $T_i(n) = 0$ for all $n > 3i-2$.

To determine the values of $T_i(n)$, we first rewrite $b_i(n)$ in a more compact form. By definition

$$b_i(n) := \sum_{k=0}^{n}\binom{2n-i-1-k}{n-i-1}\omega_{n-i}(k) - I_n(n) + \frac{i}{n}\binom{2n-i-1}{n-1} + i\sum_{k=1}^{n-1}\frac{1}{k}\binom{2k-i-1}{k-1} I_{n-k}(n-k).$$

The term $I_n(n)$ can be transformed into the sum $\sum_{k=0}^{n}\binom{2n-1-k}{n-1}\omega(k)$ by using the Knuth-Netto formula, and $\frac{i}{n}\binom{2n-i-1}{n-1}$ can be incorporated into the second summation by setting $k=n$.

$$b_i(n) = \sum_{k=0}^{n}\left[\binom{2n-i-1-k}{n-i-1}\omega_{n-i}(k) - \binom{2n-1-k}{n-1}\omega(k)\right] + i\sum_{k=1}^{n}\frac{1}{k}\binom{2k-i-1}{k-1} I_{n-k}(n-k).$$

Applying the Knuth-Netto formula once again, the second term can be written as

$$\sum_{k=1}^{n}\frac{1}{k}I_{n-k}(n-k)\binom{2k-i-1}{k-1}=\sum_{k=1}^{n}\frac{1}{k}\binom{2k-i-1}{k-1}\sum_{m=0}^{n-k}\binom{2n-2k-1-m}{n-k-1}\omega(m),$$

which, after interchanging the order of summation, takes the form

$$\sum_{m=0}^{n-1}\omega(m)\sum_{k=1}^{n-m}\frac{1}{k}\binom{2k-i-1}{k-1}\binom{2n-2k-1-m}{n-k-1}.$$

To align the summation indices across both main terms, we swap the dummy indices m and k, leading to the formula:

$$b_i(n)=\sum_{k=0}^{n}\left[\binom{2n-i-1-k}{n-i-1}\omega_{n-i}(k)-\binom{2n-1-k}{n-1}\omega(k)\right]+i\sum_{k=0}^{n-1}\omega(k)\sum_{m=1}^{n-k}\frac{1}{m}\binom{2m-i-1}{m-1}\binom{2n-2m-1-k}{n-m-1}.$$

Since the inner sum vanishes for $k=n$ (as it becomes an empty sum running from $m=1$ to 0), the upper bound of the second summation can be extended from $n-1$ to n:

$$b_i(n)=\sum_{k=0}^{n}\left[\binom{2n-i-1-k}{n-i-1}\omega_{n-i}(k)-\binom{2n-1-k}{n-1}\omega(k)\right]+i\sum_{k=0}^{n}\omega(k)\sum_{m=1}^{n-k}\frac{1}{m}\binom{2m-i-1}{m-1}\binom{2n-2m-1-k}{n-m-1}.$$

Collecting the terms containing $\omega(k)$ into the second summation, we obtain:

$$b_i(n)=\sum_{k=0}^{n}\binom{2n-1-k-i}{n-1-i}\omega_{n-i}(k)+\sum_{k=0}^{n}\omega(k)\left(-\binom{2n-1-k}{n-1}+\sum_{m=1}^{n-k}\frac{i}{m}\binom{2m-i-1}{m-1}\binom{2n-2m-1-k}{n-m-1}\right).$$

Applying the binomial identity of *Claim 14*, which asserts that:

$$-\binom{2n-1-k}{n-1}+\sum_{m=1}^{n-k}\frac{i}{m}\binom{2m-i-1}{m-1}\binom{2n-2m-1-k}{n-m-1}=\binom{2n-1-k-i}{n-1-i},$$

we can bring $b_i(n)$ into the concise form:

$$b_i(n)=\sum_{k=0}^{n}\binom{2n-i-1-k}{n-1-i}\left(\omega_{n-i}(k)-\omega(k)\right)$$

Moreover, because $\omega_{n-i}(k)=\omega(k)$ holds for $k\le n-i$, the terms for $k\le n-i$ vanish. Thus, we only need to sum from $k=n-i+1$:

$$b_i(n)=\sum_{k=n-i+1}^{n}\binom{2n-i-1-k}{n-1-i}\left(\omega_{n-i}(k)-\omega(k)\right)$$

It is worth noting that $b_i(n)$ measures the difference between the value $I^*_{n-i}(n)$, computed via the Knuth-Netto formula as

$$I^*_{n-i}(n)\overset{\text{(K-N)}}{=}\sum_{k=0}^{n}\binom{n-i+n-1-k}{n-1}\omega(k)=\sum_{k=0}^{n}\binom{2n-i-1-k}{n-1-i}\omega(k),$$

(which does not provide the correct result under the condition $n>n-i$), and the always correct value $I_{n-i}(n)$, computed according to the generalized Knuth-Netto formula as

$$I_{n-i}(n)\overset{\text{(g-K-N)}}{=}\sum_{k=0}^{n}\binom{n-i+n-1-k}{n-1}\omega_{n-i}(k)=\sum_{k=0}^{n}\binom{2n-i-1-k}{n-1-i}\omega_{n-i}(k),$$

that is, $b_i(n)=I_{n-i}(n)-I^*_{n-i}(n)$.

Now, the desired coefficients $T_i(n)$ can be easily determined; since

$$B_i(x)(1-x)^i=\left(\sum_{j=i}^{\infty}b_i(j)x^j\right)\left(\sum_{k=0}^{i}(-1)^k\binom{i}{k}x^k\right)=\sum_{j+k=n}(-1)^k\binom{i}{k}b_i(j)x^n=\sum_{j=i}^{n}(-1)^{n-j}\binom{i}{n-j}b_i(j)x^n,$$

it follows that

$$T_i(n)=(-1)^n\sum_{j=i}^{n}(-1)^j\binom{i}{n-j}\sum_{k=j-i+1}^{j}\binom{2j-i-1-k}{j-1-i}\left(\omega_{j-i}(k)-\omega(k)\right).$$

Shifting the summation index by substituting $j \longrightarrow j-i$, we obtain:

$$\text{(T)} \qquad T_i(n) = (-1)^{n+i} \sum_{j=\max(0,n-2i)}^{n-i} (-1)^j \binom{i}{n-j-i} \sum_{k=j+1}^{j+i} \binom{2j+i-1-k}{j-1} (\omega_j(k) - \omega(k))$$

Based on extensive computer verification, we propose the following claim:

Claim 9. For any positive integers i and j, the following identity holds:

$$\text{(A)} \qquad \sum_{k=j+1}^{j+i} \binom{2j+i-1-k}{j-1} (\omega_j(k) - \omega(k)) = \begin{cases} I_{j+1}(i-1) & j \geq i-1, \\ I_{j+1}(i-1)+1 & j = i-2. \end{cases}$$

The specific values of the differences $\sum_{k=j+1}^{j+i} \binom{2j+i-1-k}{j-1} (\omega_j(k) - \omega(k)) - I_{j+1}(i-1)$ for $1 \leq i, j \leq 18$ are shown in the table below:

$i \setminus j$	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.	12.	13.	14.	15.	16.	17.	18.
1.	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2.	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3.	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4.	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5.	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6.	−1	0	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
7.	−1	−1	2	3	1	0	0	0	0	0	0	0	0	0	0	0	0	0
8.	−1	−2	0	5	4	1	0	0	0	0	0	0	0	0	0	0	0	0
9.	−1	−3	−3	5	9	5	1	0	0	0	0	0	0	0	0	0	0	0
10.	−1	−3	−6	1	14	14	6	1	0	0	0	0	0	0	0	0	0	0
11.	0	−3	−9	−7	15	28	20	7	1	0	0	0	0	0	0	0	0	0
12.	0	−3	−11	−18	7	43	48	27	8	1	0	0	0	0	0	0	0	0
13.	0	−2	−12	−30	−14	50	91	75	35	9	1	0	0	0	0	0	0	0
14.	1	−1	−12	−41	−49	35	141	166	110	44	10	1	0	0	0	0	0	0
15.	1	0	−11	−50	−96	−18	176	307	276	154	54	11	1	0	0	0	0	0
16.	1	1	−9	−56	−150	−123	157	483	583	430	208	65	12	1	0	0	0	0
17.	1	2	−6	−58	−205	−288	29	640	1066	1013	638	273	77	13	1	0	0	0
18.	1	3	−2	−56	−255	−512	−273	668	1706	2079	1651	911	350	90	14	1	0	0

Proof of (A). We first prove the first case, where $j \geq i-1$. In the sum

$$S = \sum_{k=j+1}^{j+i} \binom{2j+i-1-k}{j-1} (\omega_j(k) - \omega(k)),$$

the summation index k satisfies $j+1 \leq k \leq j+i \leq 2j+1$. Since the conditions of *Claim* 6 hold, we can expand the difference $\omega_j(k) - \omega(k)$ to rewrite the sum as

$$S = \sum_{k=j+1}^{j+i} \binom{2j+i-1-k}{j-1} \sum_{l=0}^{k-j-1} \omega(l) = \sum_{l=0}^{i-1} \omega(l) \sum_{k=l+j+1}^{j+i} \binom{2j+i-1-k}{j-1}.$$

By *Claim 13*, the inner summation simplifies to

$$\sum_{k=l+j+1}^{j+i} \binom{2j+i-1-k}{j-1} = \binom{j+i-1-l}{j}.$$

Substituting this back into the expression for S and applying the Knuth-Netto formula, we obtain

$$S = \sum_{l=0}^{i-1} \binom{j+i-1-l}{j} \omega(l) = \sum_{l=0}^{i-1} \binom{(j+1)+(i-1)-1-l}{(j+1)-1} \omega(l) \overset{\text{(K-N)}}{=} I_{j+1}(i-1),$$

which completes the proof for the case $j \geq i-1$.

Next, we consider the second case, where $j = i - 2$. This condition implies that the upper limit of the summation becomes $j + i = 2j + 2$. To evaluate the sum, we isolate the final term corresponding to $k = 2j + 2$, where $\binom{3j+1-(2j+2)}{j-1} = \binom{j-1}{j-1} = 1$:

$$S = \sum_{k=j+1}^{2j+1} \binom{3j+1-k}{j-1} (\omega_j(k) - \omega(k)) + \omega_j(2j+2) - \omega(2j+2).$$

The summation from $k = j + 1$ to $2j + 1$ satisfies the boundary conditions of *Claim* 6, thus we can expand the difference $\omega_j(k) - \omega(k)$ to rewrite the sum as

$$\sum_{k=j+1}^{2j+1} \binom{3j+1-k}{j-1} (\omega_j(k) - \omega(k)) = \sum_{k=j+1}^{2j+1} \binom{3j+1-k}{j-1} \sum_{l=0}^{k-j-1} \omega(l).$$

Interchanging the order of summation and applying *Claim* 13 with the choice $i = j + 2$, gives:

$$\sum_{l=0}^{j} \omega(l) \sum_{k=j+1+l}^{2j+1} \binom{3j+1-k}{j-1} = \sum_{l=0}^{j} \left(\binom{2j+1-l}{j} - 1 \right) \omega(l) = \sum_{l=0}^{j} \binom{2j+1-l}{j} \omega(l) - \sum_{l=0}^{j} \omega(l).$$

According to the Knuth–Netto formula, we have

$$I_{j+1}(j+1) = \sum_{l=0}^{j+1} \binom{j+1+j+1-1-l}{j+1-1} \omega(l) = \sum_{l=0}^{j+1} \binom{2j+1-l}{j} \omega(l) = \sum_{l=0}^{j} \binom{2j+1-l}{j} \omega(l) + \omega(j+1).$$

By combining the above relations and applying *Claim* 8, we obtain exactly the desired statement:

$$S = I_{j+1}(j+1) - \omega(j+1) - \sum_{l=0}^{j} \omega(l) + \omega_j(2j+2) - \omega(2j+2) = I_{j+1}(j+1) + 1.$$

□

Our next claim has also undergone extensive computer verification.

Claim 10. For any integers $i \geq 2$ and $n \geq 3i - 2$, the following relation holds:

(B)
$$\sum_{j=n-2i}^{n-i} (-1)^j \binom{i}{n-j-i} I_{j+1}(i-1) = 0.$$

The table below contains the specific values of the sum $\sum_{j=n-2i}^{n-i} (-1)^j \binom{i}{n-j-i} I_{j+1}(i-1)$ for $1 \leq i \leq 8$ and $1 \leq n \leq 23$. For each row i, the column corresponding to $n = 3i - 2$ is highlighted in red:

$i \setminus n$	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.	12.	13.	14.	15.	16.	17.	18.	19.	20.	21.	22.	23.
1.	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2.	0	0	−1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3.	0	0	0	0	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4.	0	0	0	0	0	1	−2	−3	−1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5.	0	0	0	0	0	0	0	−5	−5	1	3	1	0	0	0	0	0	0	0	0	0	0	0
6.	0	0	0	0	0	0	0	0	−3	4	16	13	1	−3	−1	0	0	0	0	0	0	0	0
7.	0	0	0	0	0	0	0	0	0	−1	13	29	14	−14	−16	−2	3	1	0	0	0	0	0
8.	0	0	0	0	0	0	0	0	0	0	0	15	19	−29	−77	−51	5	18	3	−3	−1	0	0

Proof of (B). According to the generalized Knuth-Netto formula and the identity $\omega_n(k) = \mathrm{Z}_k(-\sigma_n(1), -\sigma_n(2), \ldots, -\sigma_n(k))$, we have:

$$I_{j+1}(i-1) = \sum_{m=0}^{i-1} \binom{j+1+i-1-1-m}{j+1-1} \omega_{j+1}(m) = \sum_{m=0}^{i-1} \binom{j+i-1-m}{j} \mathrm{Z}_m(-\sigma_{j+1}(1), -\sigma_{j+1}(2), \ldots, -\sigma_{j+1}(k))$$

Substituting this into the left-hand side of statement (B) , we obtain:

$$\sum_{j=n-2i}^{n-i} (-1)^j \binom{i}{n-j-i} \sum_{m=0}^{i-1} \binom{j+i-1-m}{j} \mathrm{Z}_m(-\sigma_{j+1}(1), -\sigma_{j+1}(2), \ldots, -\sigma_{j+1}(k)).$$

Since $n \geq 3i-2$, it follows that $j+1 \geq n-2i+1 \geq 3i-2-2i+1 = i-1 \geq m$. Consequently, for any index $0 \leq m \leq i-1$ in the summation, we can write

$$\mathrm{Z}_m(-\sigma_{j+1}(1), -\sigma_{j+1}(2), \ldots, -\sigma_{j+1}(k)) = \mathrm{Z}_m(-\sigma(1), -\sigma(2), \ldots, -\sigma(k)).$$

Due to the independence of the indices j and m, the order of summation can be reversed, giving

$$\sum_{m=0}^{i-1} \mathrm{Z}_m(-\sigma(1), -\sigma(2), \ldots, -\sigma(k)) \sum_{j=n-2i}^{n-i} (-1)^j \binom{i}{n-j-i} \binom{j+i-1-m}{j}.$$

By *Claim 12,* which asserts that:

$$\sum_{j=0}^{n-i} (-1)^j \binom{i}{n-j-i} \binom{j+i-1-m}{j} = \binom{m}{n-i},$$

this expression can be reduced to the simpler form:

$$\sum_{m=0}^{i-1} \binom{m}{n-i} \mathrm{Z}_m(-\sigma(1), -\sigma(2), \ldots, -\sigma(k)).$$

Since $i > 1$, we find that $n-i \geq 3i-2-i = 2i-2 > i-1 \geq m$, which implies $\binom{m}{n-i} = 0$. Therefore, every binomial coefficient in the summation vanishes, causing the entire sum to equal zero. □

We now present the main result of this chapter:

Theorem 11. *Statements (A) and (B) together imply (C).*

Proof. Let $n = 3i-2$. Since $i \geq 2$, we have $j \geq \max(0, n-2i) = \max(0, i-2) = i-2$. Therefore, we can write that

$$T_i(n) \overset{(A)}{=} (-1)^{n+i} \sum_{j=n-2i}^{n-2i} (-1)^j \binom{i}{n-j-i} \left(I_{j+1}(i-1)+1\right) + (-1)^{n+i} \sum_{j>n-2i}^{n-i} (-1)^j \binom{i}{n-j-i} I_{j+1}(i-1) =$$

$$= (-1)^{n+i+i-2} \binom{i}{3i-2-(i-2)-i} + (-1)^{n+i} \sum_{j=n-2i}^{n-i} (-1)^j \binom{i}{n-j-i} I_{j+1}(i-1) \overset{(B)}{=} (-1)^n \binom{i}{i} + 0 = (-1)^n.$$

This completes the proof that $T_i(3i-2) = (-1)^i$. On the other hand, if $n > 3i-2$, then $j > i-2$, and thus

$$T_i(n) \overset{(A)}{=} (-1)^{n+i} \sum_{j=n-2i}^{n-i} (-1)^j \binom{i}{n-j-i} I_{j+1}(i-1) \overset{(B)}{=} 0.$$

□

Combinatorial Lemmas and Proofs

Claim 12. For any non-negative integers n, m and i, the following binomial identity holds:

$$\sum_{j=0}^{n-i} (-1)^j \binom{i}{n-j-i} \binom{j+i-1-m}{j} = \binom{m}{n-i}$$

Proof. We begin by simplifying the second binomial coefficient in the summand. Using the identity of upper negation, which states that for any integer r and non-negative integer j,

$$\binom{r}{j} = (-1)^j \binom{j-r-1}{j},$$

we substitute $r = j+i-1-m$. This yields the upper term: $j-r-1 = j-(j+i-1-m)-1 = m-i$. Consequently, the second binomial coefficient can be rewritten as:

$$\binom{j+i-1-m}{j} = (-1)^j \binom{m-i}{j}.$$

Substituting this expression back into the original summation gives:

$$\sum_{j=0}^{n-i} (-1)^j \binom{i}{n-j-i} \left[(-1)^j \binom{m-i}{j}\right].$$

Since $(-1)^j \cdot (-1)^j = (-1)^{2j} = 1$ for all integers j, the alternating sign terms cancel out completely. The summation simplifies to:

$$\sum_{j=0}^{n-i} \binom{i}{n-i-j} \binom{m-i}{j}.$$

This expression is a direct application of Vandermonde's Convolution Identity, which generally states that:

$$\sum_{j} \binom{A}{K-j} \binom{B}{j} = \binom{A+B}{K}.$$

By mapping the parameters to $A = i$, $B = m - i$, and $K = n - i$, the summation evaluates to:

$$\binom{A+B}{K} = \binom{i+(m-i)}{n-i} = \binom{m}{n-i}.$$

□

Claim 13. Let i, j, and l be arbitrary non-negative integers. Then

$$\sum_{k=l+j+1}^{j+i} \binom{2j+i-1-k}{j-1} = \binom{j+i-1-l}{j}.$$

Proof. We prove the identity by using a telescoping sum based on Pascal's recurrence. By introducing a new summation index $m = 2j+i-1-k$, the limits transform such that m runs from $j-1$ up to $j+i-2-l$. Reversing the order of summation yields

$$\sum_{k=l+j+1}^{j+i} \binom{2j+i-1-k}{j-1} = \sum_{m=j-1}^{j+i-2-l} \binom{m}{j-1}.$$

By Pascal's identity, each binomial coefficient in the summand can be written as the difference of two consecutive terms:

$$\binom{m}{j-1} = \binom{m+1}{j} - \binom{m}{j}.$$

Substituting this into the rewritten summation produces a standard telescoping series:

$$\sum_{m=j-1}^{j+i-2-l} \left[\binom{m+1}{j} - \binom{m}{j} \right] = \binom{j+i-1-l}{j} - \binom{j-1}{j}.$$

Since$j \geq 1$ in the non-trivial cases and $\binom{j-1}{j} = 0$ by definition, the lower boundary term vanishes, leaving exactly $\binom{j+i-1-l}{j}$, which completes the proof □

Claim 14. Let n, i, and k be integers such that $1 \leq i \leq n$, and $0 \leq k \leq n$. Then the following binomial identity holds:

$$-\binom{2n-1-k}{n-1} + \sum_{m=1}^{n-k} \frac{i}{m} \binom{2m-i-1}{m-1} \binom{2n-2m-1-k}{n-m-1} = \binom{2n-1-k-i}{n-1-i}.$$

Proof. To prove the required binomial identity, we first rearrange the equation by moving the first term to the right-hand side. Thus, it suffices to demonstrate that

$$\sum_{m=1}^{n-k} \frac{i}{m} \binom{2m-i-1}{m-1} \binom{2n-2m-1-k}{n-m-1} = \binom{2n-1-k-i}{n-1-i} + \binom{2n-1-k}{n-1}.$$

Using the identity $\frac{a}{b}\binom{a-1}{b-1} = \binom{a}{b}$, the general term of the summation can be rewritten as

$$f(m) = \frac{i}{2m-i} \binom{2m-i}{m} \binom{2n-2m-1-k}{n-m-1}.$$

By evaluating this expression formally at $m = 0$, we find that the fictitious term corresponds precisely to the negative of the second binomial coefficient on the right-hand side:

$$f(0) = \frac{i}{-i} \binom{-i}{0} \binom{2n-1-k}{n-1} = -\binom{2n-1-k}{n-1}.$$

Consequently, extending the lower limit of the summation to $m = 0$ incorporates this term directly into the sum. The identity to be proven is therefore equivalent to the closed-form evaluation of the full sum:

$$\sum_{m=0}^{n-k} \frac{i}{2m-i} \binom{2m-i}{m} \binom{2n-2m-1-k}{n-m-1} = \binom{2n-1-k-i}{n-1-i}.$$

This expression can be recognized as a direct application of the generalized Hagen-Rothe convolution identity. To apply the identity rigorously, we recall the standard form of the Hagen-Rothe convolution:

$$\sum_{m=0}^{M} \frac{a}{a+b\cdot m}\binom{a+b\cdot m}{m}\binom{c-b\cdot m}{M-m} = \binom{a+c}{M}.$$

Let us set $M = n-k$. By using the symmetry property of binomial coefficients, $\binom{N}{K} = \binom{N}{N-K}$, the second binomial coefficient in our summation can be rewritten as

$$\binom{2n-2m-1-k}{n-m-1} = \binom{2n-k-1-2m}{n-m}.$$

Substituting this back into the full sum, we obtain

$$\sum_{m=0}^{n-k} \frac{i}{2m-i}\binom{2m-i}{m}\binom{2n-k-1-2m}{n-m}.$$

We can now map this expression directly onto the Hagen-Rothe [1] identity by choosing the parameters $a = -i$, $b = 2$, $c = 2n-k-1$, and keeping the upper limit as $M = n-k$. Note that the factor $\frac{i}{2m-i}$ matches $\frac{-i}{-i+2m}$ exactly.

The convolution theorem then states that the sum evaluates to a single binomial coefficient:

$$\binom{a+c}{M} = \binom{-i+2n-k-1}{n-k} = \binom{2n-k-i-1}{n-k}.$$

Finally, applying the symmetry property $\binom{N}{K} = \binom{N}{N-K}$ once more to the result, we find

$$\binom{2n-k-i-1}{(2n-k-i-1)-(n-k)} = \binom{2n-1-k-i}{n-1-i},$$

which completes the proof of the identity. □

The polynomials $P_{i,A}(x)$, $P_{i,B}(x)$ és $P_i(x)$ for some small values of $i \geq 2$.

For any positive integer i, the evaluations $P_{i,A}(1) = 0$, $P_{i,B}(1) = 1$, and $P_i(1) = 1$ hold. The table below presents the polynomials $P_{i,A}(x)$, $P_{i,B}(x)$, and $P_i(x)$ for values of i ranging from 2 to 6.

Table 6. Polynomials $P_{i,A}(x)$, $P_{i,B}(x)$, and $P_i(x)$.

i		
2	$P_{2,A}$	$-1+4\,x-5\,x^2+2\,x^3$
	$P_{2,B}$	$x^2-x^3+x^4$
	P_2	$-1+4\,x-4\,x^2+x^3+x^4$
3	$P_{3,A}$	$-1+6\,x-12\,x^2+10\,x^3-3\,x^4$
	$P_{3,B}$	$x^4-x^5+2\,x^6-x^7$
	P_3	$-1+6\,x-12\,x^2+10\,x^3-2\,x^4-x^5+2\,x^6-x^7$
4	$P_{4,A}$	$-1+8\,x-24\,x^2+35\,x^3-25\,x^4+6\,x^5+2\,x^6-x^7$
	$P_{4,B}$	$2\,x^6-2\,x^7+3\,x^8-3\,x^9+x^{10}$
	P_4	$-1+8\,x-24\,x^2+35\,x^3-25\,x^4+6\,x^5+4\,x^6-3\,x^7+3\,x^8-3\,x^9+x^{10}$
5	$P_{5,A}$	$-1+10\,x-40\,x^2+84\,x^3-110\,x^4+116\,x^5-120\,x^6+100\,x^7-49\,x^8+10\,x^9$
	$P_{5,B}$	$-x^5+5\,x^6-9\,x^7+11\,x^8-5\,x^9-2\,x^{11}+3\,x^{12}-x^{13}$
	P_5	$-1+10\,x-40\,x^2+84\,x^3-110\,x^4+115\,x^5-115\,x^6+91\,x^7-38\,x^8+5\,x^9-2\,x^{11}+3\,x^{12}-x^{13}$
6	$P_{6,A}$	$-1+12\,x-60\,x^2+165\,x^3-287\,x^4+314\,x^5-99\,x^6-350\,x^7+680\,x^8-578\,x^9+247\,x^{10}-43\,x^{11}$
	$P_{6,B}$	$-x^7+6\,x^8-10\,x^9+13\,x^{10}-7\,x^{11}-6\,x^{12}+8\,x^{13}-3\,x^{15}+x^{16}$
	P_6	$-1+12\,x-60\,x^2+165\,x^3-287\,x^4+314\,x^5-99\,x^6-351\,x^7+686\,x^8-588\,x^9+260\,x^{10}-50\,x^{11}-6\,x^{12}+8\,x^{13}-3\,x^{15}+x^{16}$

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